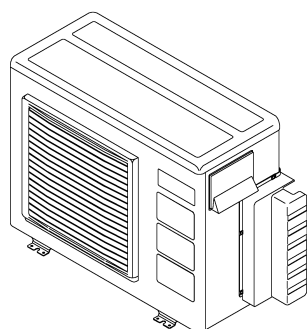
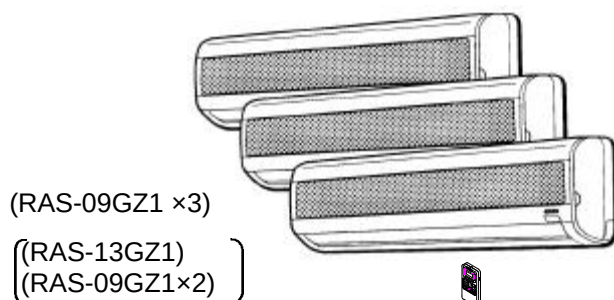


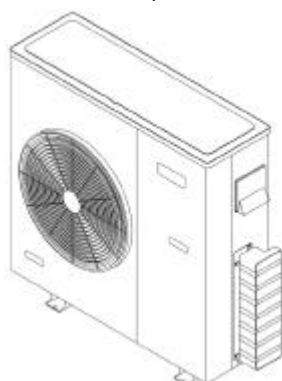
# HITACHI

## SERVICE MANUAL TECHNICAL INFORMATION

FOR SERVICE PERSONNEL ONLY



(RAM-20GW1)



(RAM-30GW1)

TH NO. S0007 E

RAS-09GZ1x3/RAM-20GW1  
RAS-13GZ1 /RAM-30GW1  
RAS-09GZ1x2

REFER TO THE FOUNDATION  
MANUAL

|                                                    |     |
|----------------------------------------------------|-----|
| SPECIFICATIONS.....                                | 7   |
| HOW TO USE.....                                    | 10  |
| CONSTRUCTION                                       | AND |
| DIMENSIONAL DIAGRAM.....                           | 21  |
| MAIN PARTS COMPONENT.....                          | 23  |
| WIRING DIAGRAM.....                                | 27  |
| WIRING DIAGRAM OF THE PRINTED<br>WIRING BOARD..... | 33  |
| BASIC MODE.....                                    | 37  |
| AUTO SWING FUNCTION.....                           | 42  |
| DESCRIPTION OF MAIN CIRCUIT OPERATION.....         | 43  |
| SERVICE CALL Q&A.....                              | 47  |
| TROUBLE SHOOTING.....                              | 50  |
| PARTS LIST AND DIAGRAM.....                        | 52  |

### SPECIFICATIONS

| TYPE             |           | WALL TYPE        |              |                          |              |
|------------------|-----------|------------------|--------------|--------------------------|--------------|
|                  |           | INDOOR UNIT      | OUTDOOR UNIT | INDOOR UNIT              | OUTDOOR UNIT |
| MODEL            |           | RAS-09GZ1x3      | RAM-20GW1    | RAS-13GZ1<br>RAS-09GZ1x2 | RAM-30GW1    |
| POWER SOURCE     |           | 1φ,220-230V,50Hz |              |                          |              |
| TOTAL INPUT      | (W)       | 1,850-1,925      |              | 3,405-3,606              |              |
| TOTAL AMPERES    | (A)       | 8.5-8.5          |              | 16.2-16.0                |              |
| COOLING CAPACITY | (kW)      | 2.6 + 1.6x2      |              | 3.0 + 2.4x2              |              |
|                  | (B.T.U/h) | 8,870 + 5460x2   |              | 10,240 + 8,190x2         |              |
| DIMENSIONS (mm)  | W         | 823              | 810          | 823                      | 890          |
|                  | H         | 298              | 610          | 298                      | 885          |
|                  | D         | 179 (185)        | 350          | 179 (185)                | 285          |
| NET WEIGHT       | (kg)      | 8x3              | 61           | 8x3                      | 80           |

After installation

SPECIFICATIONS AND PARTS ARE SUBJECT TO CHANGE FOR IMPROVEMENT

## ROOM AIR CONDITIONER

INDOOR UNIT + OUTDOOR UNIT

FEBRUARY 2001

## SAFETY DURING REPAIR WORK

1. In order to disassemble and repair the unit in question, be sure to disconnect the power cord plug from the power outlet before starting the work.
2. If it is necessary to replace any parts, they should be replaced with respective genuine parts for the unit, and the replacement must be effected in correct manner according to the instructions in the Service Manual of the unit.

|                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|
| If the contacts of electrical parts are defective, replace the electrical parts without trying to repair them. |
|----------------------------------------------------------------------------------------------------------------|

3. After completion of repairs, the initial state should be restored.
4. Lead wires should be connected and laid as in the initial state.
5. Modification of the unit by user himself should absolutely be prohibited.
6. Tools and measuring instruments for use in repairs or inspection should be accurately calibrated in advance.
7. In installing the unit having been repaired, be careful to prevent the occurrence of any accident such as electrical shock, leak of current, or bodily injury due to the drop of any part.
8. To check the insulation of the unit, measure the insulation resistance between the power cord plug and grounding terminal of the Unit. The insulation resistance should be 1 MΩ or more as measured by a 500V DC megger.
9. The initial location of installation such as window, floor or the other should be checked for being and safe enough to support the repaired unit again. If it is found not so strong and safe, the unit should be installed at the initial location reinforced or at a new location.
10. Any inflammable thing should never be placed about the location of installation.
11. Check the grounding to see whether it is proper or not, and if it is found improper, connect the grounding terminal to the earth.

## PREVENTION OF DAMAGE TO SEMICONDUCTORS

1. when carrying and handling semiconductors adopted in your model during maintenance and inspection thereof, much care should be taken to prevent the semiconductors from being damaged. Also, such care should be taken when handling any faulty model which is to be returned to factory.

2. The semiconductors used in your model are the following:

- (1) Micro computer
- (2) integrated circuits (IC)
- (3) Field-effect transistors (FET)
- (4) Printed circuit boards (PC boards) or the like on which the parts in (1) and (2) above are provided.

3. Cautions in handling

- (1) Use a conductive container to carry or store the semiconductive parts. Even if they are faulty ones, also handle them using such container.

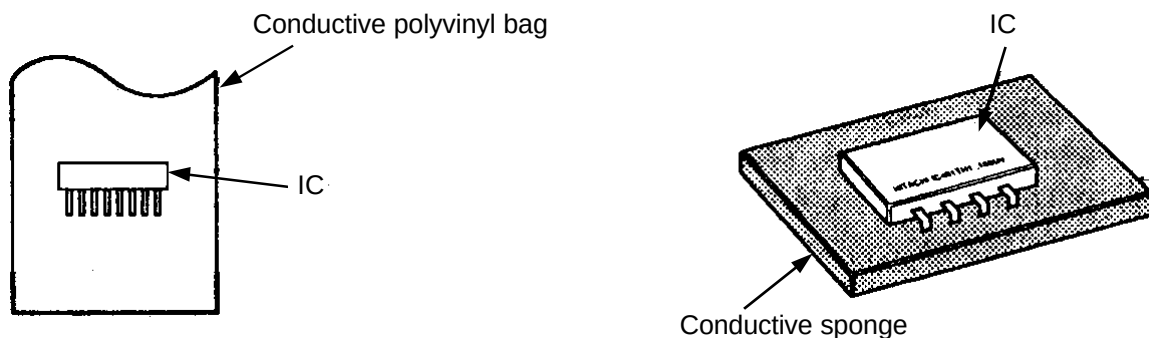


Fig. 1. Conductive Container

- (2) when parts as uncovered are handled (for counting, packing or for the like purpose), the handler must use his own body as conductor for earthing. For this purpose, put on an electrically conductive ring or bracelet at the wrist. Connect to the bracelet a conductor provided with a resistor of  $1M\Omega$  and at the other end with a clip for connection to the earth wire.

- (3) Be careful not to have your clothes be in contact with any part while you are holding it, even if the body earthing is established.
- (4) Be sure to place the parts on a grounded metallic plate.
- (5) Never fail to disconnect the power supply before starting repair of any PC board. Then, proceed to the repair of the PC board on the grounded metallic plate.

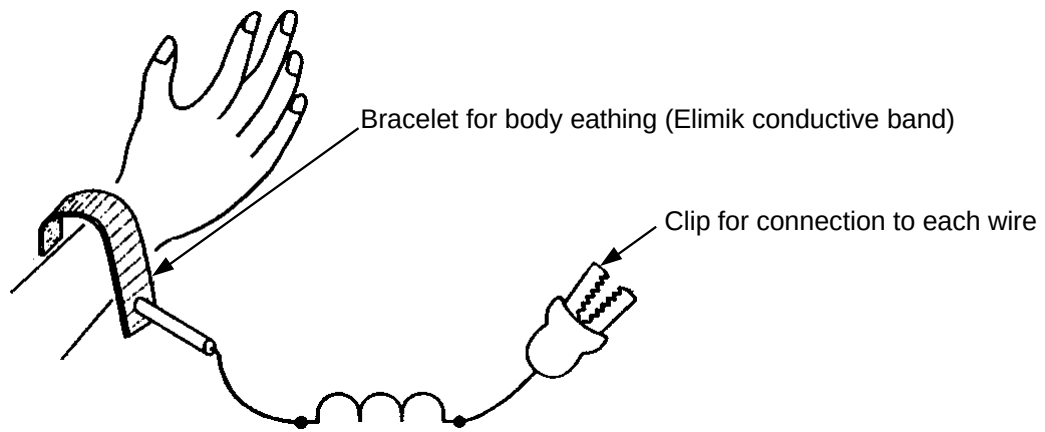


Fig. 2. Body Earthing

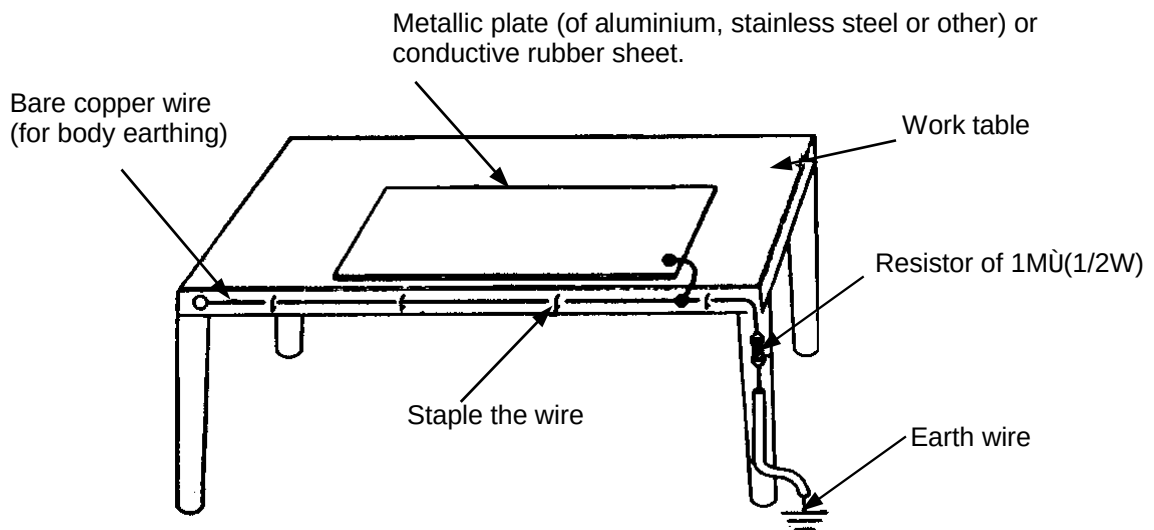


Fig. 3. Earthing of Work Table

- (6) Soldering iron to be used should be a one with three wire (including an earth wire).

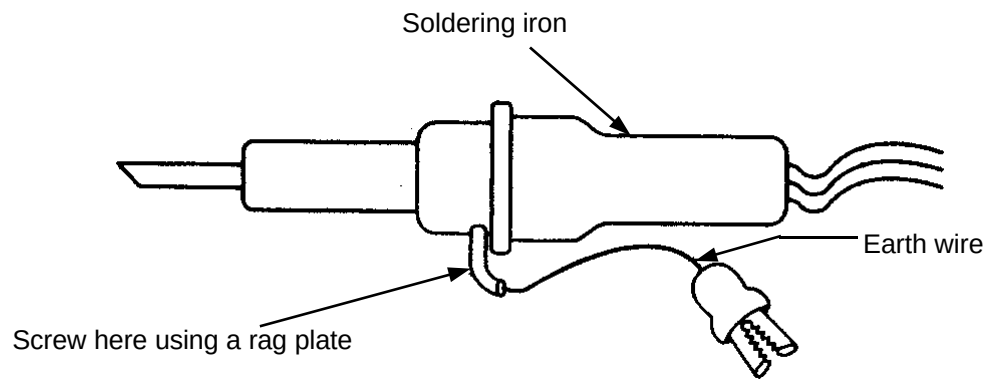


Fig.4.Earthing of Soldering Iron

An ordinary soldering iron may also be used, but in such case, be sure to provide a perfect insulation (10M $\Omega$  or more to 100V).

- (7) while checking the circuits during maintenance, inspection or the other, strictly avoid any shortcircuiting of the load circuit or other by the test probe of the measuring instrument.

## NEW TECHNOLOGY

### Cautions in relation to HFC(R407C)

#### 1. Safety during Servicing

This air conditioner uses the new refrigerant HFC(R407C) for protecting the ozone layer. R407C has several different characteristic features from HCFC22(R22). Therefore keep the following care items during servicing for safety.

- (1) Since the working pressure of R407C model is about 1.1 times higher than that of R22 model, it becomes necessary to use part of piping materials and servicing tools exclusive for R407C model.
- (2) It is necessary to exercise more care to prevent the foreign matters (oil, moisture, etc.) from mixing into the piping than in the case of R22 model. Also, when storing the piping securely seal its openings with pinching and taping, etc..
- (3) Be sure to charge the refrigerant from the liquid-phase side, as the liquid-phase/gas-phase-composition changes a little in the case of R407C model.
- (4) Never use refrigerant other than R407C in an air conditioner which is designed to operate with R407C.
- (5) If a refrigeration gas leakage occurs during servicing, be sure to ventilate fully.

If the refrigerant gas comes into contact with fire, a poisonous gas may occur.

- (6) When installing or removing an air conditioner, do not allow air or moisture to remain in the refrigeration cycle. Otherwise, pressure in the refrigeration cycle may become abnormally high so that a rupture or personal injury may be caused.

- (7) After completion of service work, check to make sure that there is no refrigeration gas leakage.

If the refrigerant gas leaks into the room, coming into contact with fire in the fan driven heater, space heater, etc., a poisonous gas may occur.

- (8) When exchange compressor, it is necessary to change the dryer at the same time.

#### 2.Refrigerant Piping Materials

Thickness of Refrigerant Piping  
Although the thickness is same  
as that for R22

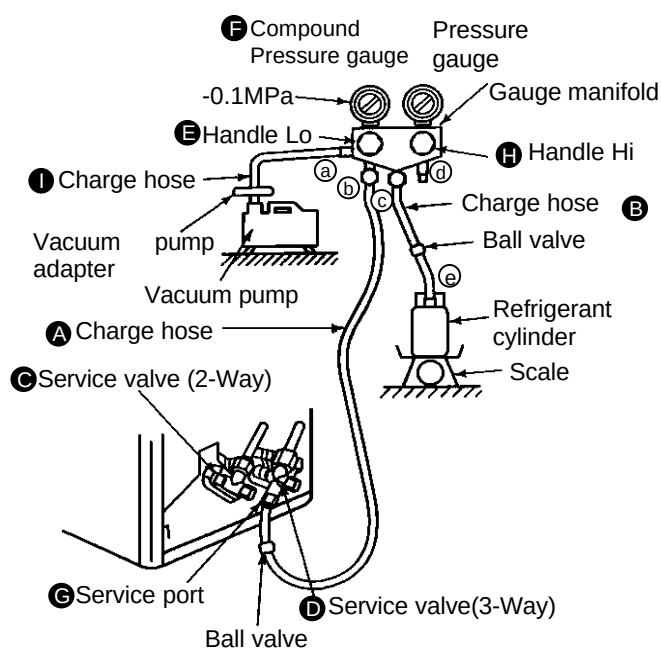
| Nominal | outside (mm) | thickness(mm) |
|---------|--------------|---------------|
| 1/4     | 6.35         | 0.7           |
| 3/8     | 9.52         | 0.7           |
| 1/2     | 12.70        | 0.7           |

#### 3. Servicing tools

<New tools for R407C>

| New tools for R407C                    | Applicable to R22 Model | Applicable to R410A Model  | Changes                                                                                                                                              |
|----------------------------------------|-------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gauge manifold                         | ×                       | ×                          | Use tools for R407C when port dia. is 1/4 flare.                                                                                                     |
| Charge hose                            | ×                       | ×                          |                                                                                                                                                      |
| Weight measure before R407C be charged |                         |                            | As pressure is high and gasification speed is fast, it is difficult to read the indicated value by means of charging cylinder, as air bubbles occur. |
| Torque wrench                          |                         | × (nominal diam. 1/2, 5/8) | Can share the tool for R22                                                                                                                           |
| Flare tool (clutch type)               |                         |                            |                                                                                                                                                      |
| Gauge for projection adjustment        |                         |                            | Used when performing flare processing by means of conventional flare tool.                                                                           |
| Vacuum pump adapter                    |                         |                            | Connected to conventional vacuum pump.                                                                                                               |
| Gas leakage detector                   | ×                       |                            | Exclusive for HFC refrigerant.                                                                                                                       |

#### 4. Servicing work (Refrigerant recharging)



**CAUTION**

1. Be sure to use the vacuum pump, vacuum pump adapter and gauge manifold to refer to their instruction manuals beforehand.
2. Ascertain that the vacuum pump is filled with oil to the level designated on the oil gauge.
3. After closed the ball valve of change hose, it should be disconnected at service port side and refrigerant cylinder side at first.

Next, after discharging the remained gas in the charge hose by opening the ball valve a little, disconnect it at gauge manifold side. You can prevent from being released the refrigerant suddenly by connecting the ball valve to service port. And you can work with more safety.

## Working steps

- 
- ```

graph TD
    1[1. Connect the charge hose (A) to outdoor unit.] --> 2[2. Connect the vacuum pump adapter to the vacuum pump. Connect the 1/2" conversion adapter to the vacuum pump adapter. Connect the charge hose (I) to the conversion adapter.]
    2 --> 3[3. Connect the charge hose (B) to the refrigerant cylinder.]
    3 --> 4[4. Open the handle Lo (E)  
Turn ON the power switch of the vacuum pump & adapter.  
Run the vacuum pump in specified time.]
    4 --> 5[5. When the compound pressure gauge's pointer has indicated -0.1 Mpa, place the handle (E) in the fully closed position.  
Turn OFF the power switch of the vacuum pump & adapter.]
    5 --> 6[6. Remove the charge hose (I) of vacuum pump at portion (a)]
    6 --> 7[7. Air purge of gauge manifold.  
● Open the refrigerant cylinder's valve and push the valve core at portion (a) of gauge manifold. Then the refrigerant is discharge in a moment.]
    7 --> 8[8. Calculation of charged refrigerant amount.]
    8 --> 9[9. Charging of refrigeration.  
● Open the handle Lo (E) in a turn and charge the designated amount.]
    9 --> 10[10. Completion of charging.]
    10 --> 11[11. Be closed the valve of charge hose (A)]
    11 --> 12[12. Run the compressor at cooling operation.]
    12 --> 13[13. Remove the charge hose (A & B)  
● Remove the charge hose (A) from portion (G)  
● Remove the charge hose (B) from portion (e)]
    13 --> 14[14. Attach the caps.]
    14 --> 15[15. Gas leakage check.]
  
```
1. Connect the charge hose (A) to outdoor unit.
  2. Connect the vacuum pump adapter to the vacuum pump. Connect the 1/2" conversion adapter to the vacuum pump adapter. Connect the charge hose (I) to the conversion adapter.
  3. Connect the charge hose (B) to the refrigerant cylinder.
  4. Open the handle Lo (E)
 

Turn ON the power switch of the vacuum pump & adapter.

 Run the vacuum pump in specified time.
  5. When the compound pressure gauge's pointer has indicated -0.1 Mpa, place the handle (E) in the fully closed position.
 

Turn OFF the power switch of the vacuum pump & adapter.
  6. Remove the charge hose (I) of vacuum pump at portion (a)
  7. Air purge of gauge manifold.
    - Open the refrigerant cylinder's valve and push the valve core at portion (a) of gauge manifold. Then the refrigerant is discharge in a moment.
  8. Calculation of charged refrigerant amount.
  9. Charging of refrigeration.
    - Open the handle Lo (E) in a turn and charge the designated amount.
  10. Completion of charging.
  11. Be closed the valve of charge hose (A)
  12. Run the compressor at cooling operation.
  13. Remove the charge hose (A & B)
    - Remove the charge hose (A) from portion (G)
    - Remove the charge hose (B) from portion (e)
  14. Attach the caps.
  15. Gas leakage check.

## 5. Saturated Temperature and Saturated Pressure table of R407C model

| Temperature<br>( ) | Saturated pressure (MPa) |               |
|--------------------|--------------------------|---------------|
|                    | Saturated liquid         | Saturated gas |
| -10                | 0.3063                   | 0.2182        |
| -9                 | 0.3208                   | 0.2305        |
| -8                 | 0.3357                   | 0.2431        |
| -7                 | 0.3510                   | 0.2562        |
| -6                 | 0.3667                   | 0.2696        |
| -5                 | 0.3828                   | 0.2834        |
| -4                 | 0.3993                   | 0.2977        |
| -3                 | 0.4163                   | 0.3123        |
| -2                 | 0.4337                   | 0.3273        |
| -1                 | 0.4515                   | 0.3427        |
| 0                  | 0.4698                   | 0.3586        |
| 1                  | 0.4885                   | 0.3749        |
| 2                  | 0.5077                   | 0.3916        |
| 3                  | 0.5274                   | 0.4088        |
| 4                  | 0.5475                   | 0.4265        |
| 5                  | 0.5681                   | 0.4446        |
| 6                  | 0.5892                   | 0.4632        |
| 7                  | 0.6108                   | 0.4822        |
| 8                  | 0.6329                   | 0.5018        |
| 9                  | 0.6556                   | 0.5219        |
| 10                 | 0.6787                   | 0.5424        |
| 11                 | 0.7024                   | 0.5635        |
| 12                 | 0.7266                   | 0.5851        |
| 13                 | 0.7513                   | 0.6073        |
| 14                 | 0.7766                   | 0.6299        |
| 15                 | 0.8025                   | 0.6532        |
| 16                 | 0.8289                   | 0.6770        |
| 17                 | 0.8559                   | 0.7013        |

| Temperature<br>( ) | Saturated pressure (MPa) |               |
|--------------------|--------------------------|---------------|
|                    | Saturated liquid         | Saturated gas |
| 18                 | 0.8835                   | 0.7263        |
| 19                 | 0.9117                   | 0.7518        |
| 20                 | 0.9407                   | 0.7779        |
| 21                 | 0.9697                   | 0.8047        |
| 22                 | 1.0000                   | 0.8320        |
| 23                 | 1.031                    | 0.8600        |
| 24                 | 1.062                    | 0.8886        |
| 25                 | 1.094                    | 0.9177        |
| 26                 | 1.126                    | 0.9477        |
| 27                 | 1.159                    | 0.9787        |
| 28                 | 1.193                    | 1.010         |
| 29                 | 1.228                    | 1.032         |
| 30                 | 1.263                    | 1.075         |
| 31                 | 1.299                    | 1.108         |
| 32                 | 1.335                    | 1.142         |
| 33                 | 1.373                    | 1.177         |
| 34                 | 1.411                    | 1.212         |
| 35                 | 1.449                    | 1.239         |
| 36                 | 1.489                    | 1.286         |
| 37                 | 1.529                    | 1.323         |
| 38                 | 1.570                    | 1.362         |
| 39                 | 1.611                    | 1.401         |
| 40                 | 1.654                    | 1.442         |
| 41                 | 1.697                    | 1.483         |
| 42                 | 1.741                    | 1.525         |
| 43                 | 1.785                    | 1.567         |
| 44                 | 1.831                    | 1.611         |
| 45                 | 1.877                    | 1.655         |



## SPECIFICATIONS

MODEL RAS-09GZ1×3/RAM-20GW1

|                                             |                     |                                                                                                             |                       |
|---------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------|-----------------------|
| MODEL                                       |                     | RAS-09GZ1×3                                                                                                 | RAM-20GW1             |
| FAN MOTOR                                   |                     | 8W                                                                                                          | 40W                   |
| FAN MOTOR CAPACITOR                         |                     | 1μF, 450V                                                                                                   | 2.3μF, 450V           |
| FAN MOTOR PROTECTOR                         |                     | YES                                                                                                         | YES                   |
| COMPRESSOR                                  |                     | NO                                                                                                          | GK516S1Z×2            |
| COMPRESSOR MOTOR CAPACITOR                  |                     | NO                                                                                                          | 40μF, 400V            |
| OVER HEAT PROTECTOR                         |                     | NO                                                                                                          | NO                    |
| OVERLOAD RELAY                              |                     | NO                                                                                                          | YES                   |
| FUSE ( for MICRO COMPUTER )                 |                     | 2A                                                                                                          | 2.5A                  |
| FUSE                                        |                     | NO                                                                                                          | NO                    |
| POWER RELAY                                 |                     | JM1AN                                                                                                       | NO                    |
| POWER SWITCH                                |                     | YES                                                                                                         | NO                    |
| TEMPORARY SWITCH                            |                     | YES                                                                                                         | NO                    |
| SERVICE SWITCH                              |                     | NO                                                                                                          | NO                    |
| TRANSFORMER                                 |                     | YES                                                                                                         | YES                   |
| VARISTOR                                    |                     | 470NR                                                                                                       | 470NR                 |
| NOISE SUPPRESSOR                            |                     | NO                                                                                                          | NO                    |
| THERMOSTAT                                  |                     | YES ( IC )                                                                                                  | NO                    |
| SOLID STATE RELAY FOR FAN                   |                     | ALD-112                                                                                                     | NO                    |
| REMOTE CONTROLLER ( LIQUID CRYSTAL )        |                     | YES                                                                                                         | NO                    |
| FUSE CAPACITY                               |                     | 15A TIME DELAY FUSE                                                                                         |                       |
| REFRIGERANT CHARGING<br>VOLUME<br>( R407C ) | UNIT                |                                                                                                             | A : 830g<br>B : 1000g |
|                                             | REFER TO<br>Page. 9 | WITHOUT REFRIGERANT BECAUSE<br>COUPLING IS FLARE TYPE.<br>P-103VK1( 3m ), P-105VK1( 5m ),<br>P-108VK1( 8m ) |                       |
| PRODUCED BY.                                |                     | TAINICHI                                                                                                    |                       |

A : RAS-09GZ1 ( No.1 )

B : RAS-09GZ1 ( No.2 No.3 )

## SPECIFICATIONS

MODEL RAS-13GZ1 / RAM-30GW1  
RAS-09GZ1×2

|                                          |                     |                                                                                                             |             |                                  |
|------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------|-------------|----------------------------------|
| MODEL                                    |                     | RAS-13GZ1                                                                                                   | RAS-09GZ1×2 | RAM-30GW1                        |
| FAN MOTOR                                |                     | 10W                                                                                                         | 8W          | 75W                              |
| FAN MOTOR CAPACITOR                      |                     | 1μF, 450V                                                                                                   | 1μF, 450V   | 4μF, 500V                        |
| FAN MOTOR PROTECTOR                      |                     | YES                                                                                                         | YES         | YES                              |
| COMPRESSOR                               |                     | NO                                                                                                          | NO          | A : 4KS210DAA<br>B : 4KS340DAA   |
| COMPRESSOR MOTOR CAPACITOR               |                     | NO                                                                                                          | NO          | A : 35μF, 400V<br>B : 60μF, 400V |
| OVER HEAT PROTECTOR                      |                     | NO                                                                                                          | NO          | NO                               |
| OVERLOAD RELAY                           |                     | NO                                                                                                          | NO          | YES                              |
| FUSE( for MICRO COMPUTER )               |                     | 2A                                                                                                          | 2A          | 2.5A                             |
| FUSE                                     |                     | NO                                                                                                          | NO          | NO                               |
| POWER RELAY                              |                     | JM1aN                                                                                                       | JM1aN       | NO                               |
| POWER SWITCH                             |                     | YES                                                                                                         | YES         | NO                               |
| TEMPORARY SWITCH                         |                     | YES                                                                                                         | YES         | NO                               |
| SERVICE SWITCH                           |                     | NO                                                                                                          | NO          | NO                               |
| TRANSFORMER                              |                     | YES                                                                                                         | YES         | YES                              |
| VARISTOR                                 |                     | 470NR                                                                                                       | 470NR       | 470NR                            |
| NOISE SUPPRESSOR                         |                     | NO                                                                                                          | NO          | NO                               |
| THERMOSTAT                               |                     | YES ( IC )                                                                                                  | YES ( IC )  | NO                               |
| SOLID STATE RELAY FOR FAN                |                     | ALD-112                                                                                                     | ALD-112     | NO                               |
| REMOTE CONTROLLER ( LIQUID CRYSTAL )     |                     | YES                                                                                                         | YES         | NO                               |
| FUSE CAPACITY                            |                     | 25A TIME DELAY FUSE                                                                                         |             |                                  |
| REFRIGERANT CHARGING<br>VOLUME ( R407C ) | UNIT                |                                                                                                             |             | A : 850g<br>B : 1070g            |
|                                          | REFER TO<br>Page. 9 | WITHOUT REFRIGERANT BECAUSE COUPLING<br>IS FLARE TYPE.<br>P-103VK1( 3m ), P-105VK1( 5m ),<br>P-108VK1( 8m ) |             |                                  |
| PRODUCED BY.                             |                     | TAINICHI                                                                                                    |             |                                  |

A : RAS-13GZ1 ( No.1 )  
B : RAS-09GZ1 ( No.2 No.3 )

## PIPE LENGTH AND REFRIGERANT CHARGING VOLUME

1. Compressor-1 Indoor unit cycle (RAS-09GZ1/RAM-20GW1 or RAS-13GZ1/RAM-30GW1)

• The pipe length should be a maximum of 20 m.

• Charging with refrigerant is not required for a pipe with a length of up to 8m. When the pipe length exceeds 8m, charge with 10g of refrigerant for each 1 m over 8 m.

| PIPE LENGTH BETWEEN THE INDOOR UNIT AND OUTDOOR UNIT. | CHARGING VOLUME                                                             |
|-------------------------------------------------------|-----------------------------------------------------------------------------|
| UP TO TOTAL LENGTH 8 m.                               | PRECHARGING VOLUME TO OUTDOOR UNIT.                                         |
| EXCEEDS 8 m (UP TO TOTAL LENGTH 20 m).                | ADD EXTRA REFRIGERANT IN RATIO OF 10g FOR EACH 1 m IN EXCESS 8 m(See Fig.5) |

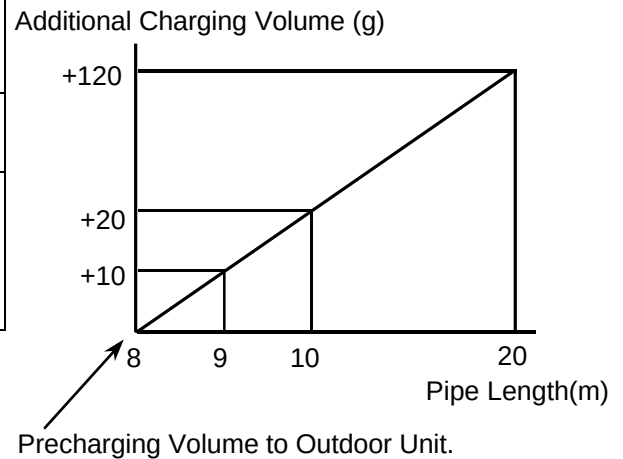


Fig.5

2. Compressor-2 Indoor unit cycle (RAS-09GZ1×2/RAM-20GW1 or RAS-09GZ1×2/RAM-30GW1)

• The total pipe length when connecting 2 indoor units should be a maximum of 20 m.

<Example>

|             | RAS-09GZ1 ( No.2 ) | RAS-09GZ1 ( No.3 ) | Judgement  |
|-------------|--------------------|--------------------|------------|
| Pipe length | 5 m                | 15 m               | Acceptable |
|             | 10 m               | 10 m               | Acceptable |
|             | 3 m                | 17 m               | Rejected   |

• In any case, charging with refrigerant is not required.

Note :

• If refrigerant recharge is required during service, be sure to turn on the unit when refrigerant discharging-vacuum drawing-refrigerant recharging work to ensure an accurate vacuum degree because there is a solenoid check valve at pipe system.

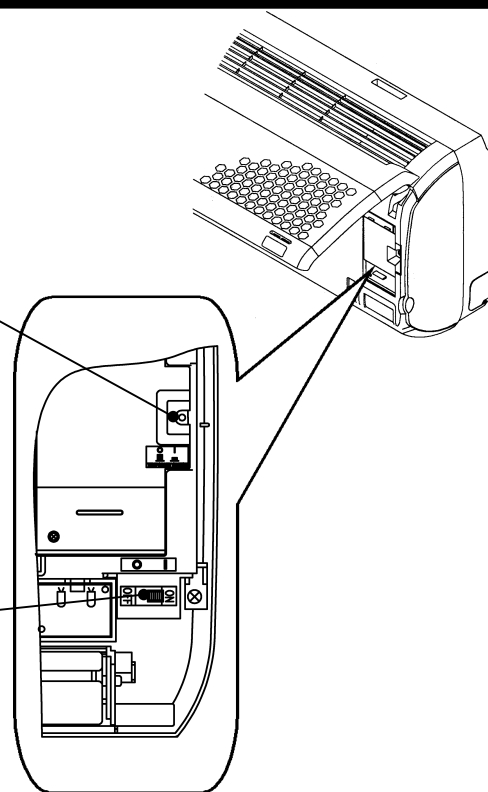
## HOW TO USE INDOOR UNIT CONTROL PANEL

- Hold both sides of the suction grille and lift it up to open it.
- After the work is finished, slightly lift the suction grille and then close it.

- **TEMPORARY SWITCH WITH AUTO RESTART**

Use this switch to start and stop when the remote controller does not work, or when the operation is done using the temporary switch with auto restart after the power source is turned off and is turned on again, the operation is done in automatic mode. Normally do not use this button.

- **POWER SWITCH**

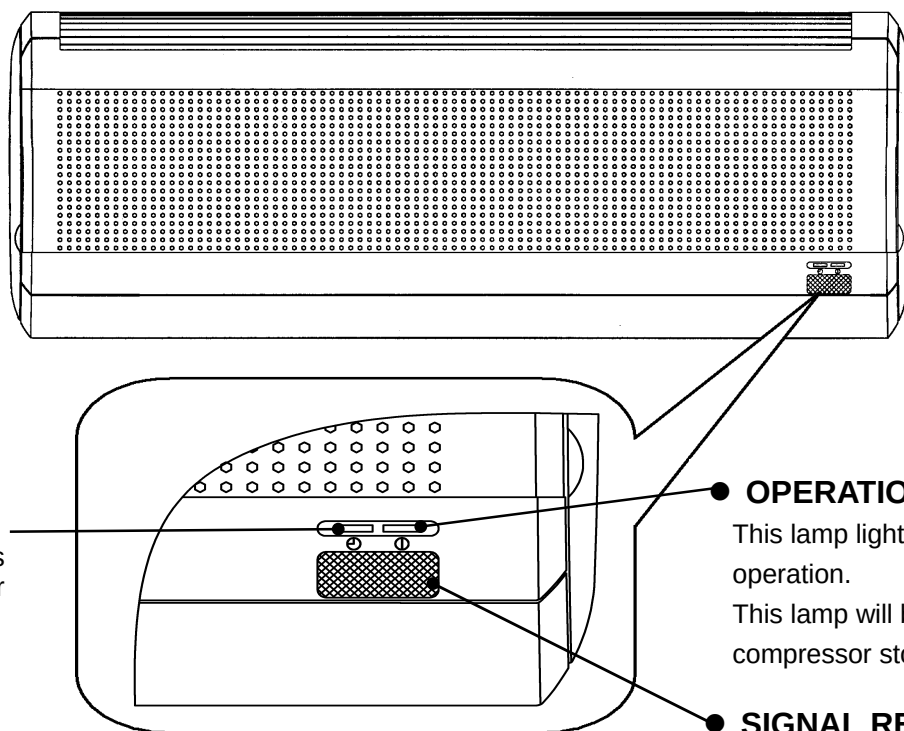


## INDOOR UNIT INDICATORS

- **TIMER lamp**  
This lamp lights when the timer is working.

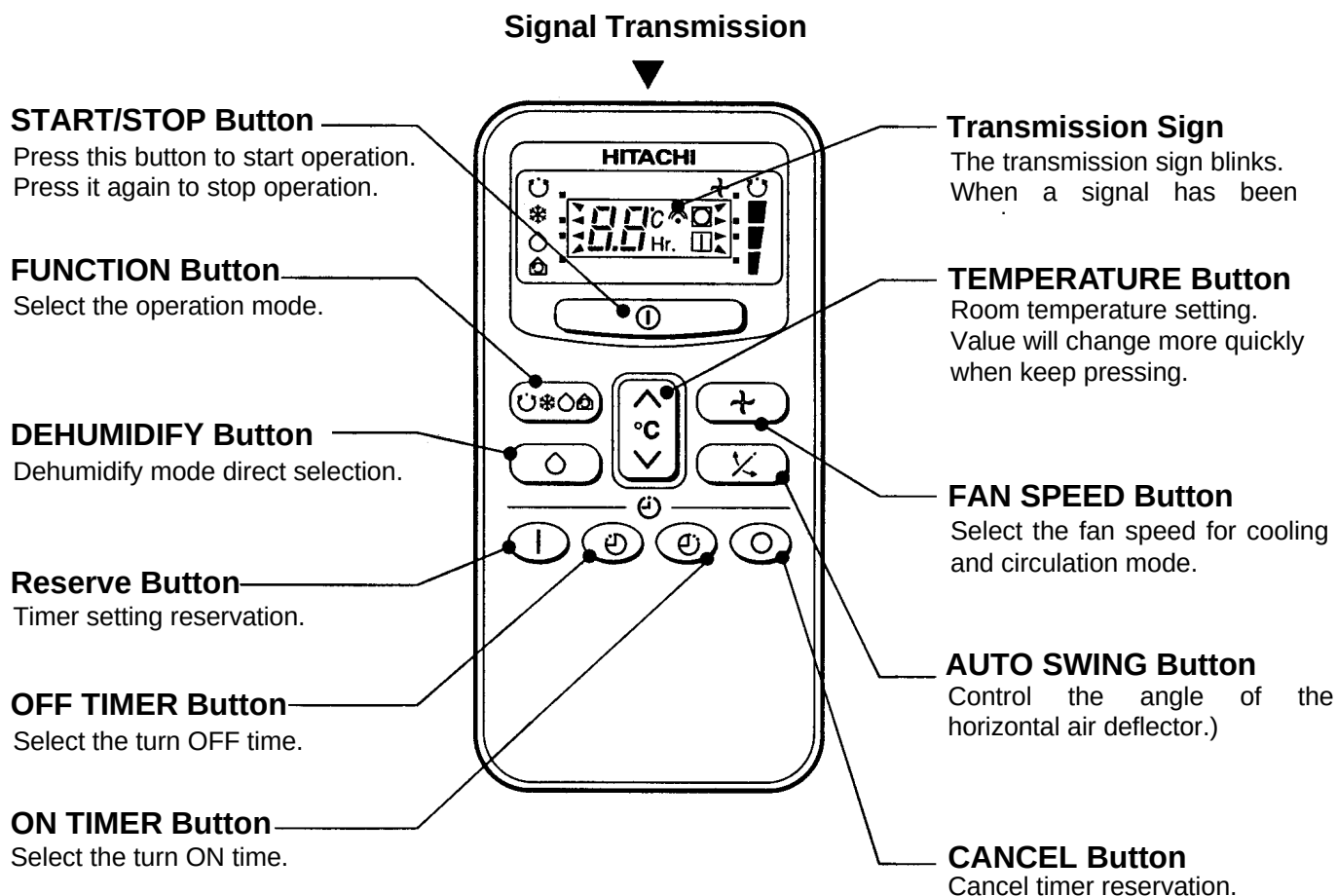
- **OPERATION lamp**  
This lamp lights during operation.  
This lamp will blink when compressor stops operating

- **SIGNAL RECEIVER**  
There will be a beep sound when this receiver receive signal from remote controller.



## NAMES AND FUNCTIONS OF REMOTE CONTROLLER

This controls the operation function and timer setting of the room air conditioner. The range of control is about 4 meters. If indoor lighting is controlled electronically, the range of control may be shorter.



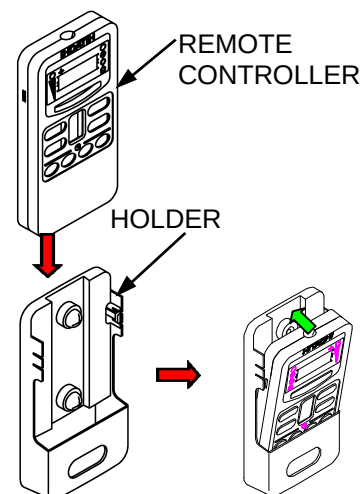
|  |                 |
|--|-----------------|
|  | AUTO            |
|  | COOL            |
|  | DEHUMIDIFY      |
|  | CIRCULATION     |
|  | START/STOP      |
|  | FAN             |
|  | AUTO SWING      |
|  | START (RESERVE) |
|  | OFF TIMER SET   |
|  | ON TIMER SET    |
|  | STOP (CANCEL)   |
|  | TIMER SET       |
|  | FAN SPEED       |
|  | HI              |
|  | MED             |
|  | LOW             |

### Precautions for Use

- Do not put the remote controller in direct sunlight and high temperature.
- Do not drop it on the floor, and protect it from water.
- If you press the **FUNCTION** selector button during operation, the air conditioner may stop for about 3 minutes for protection before you can start it again. (unless you turn off and on again the power source)

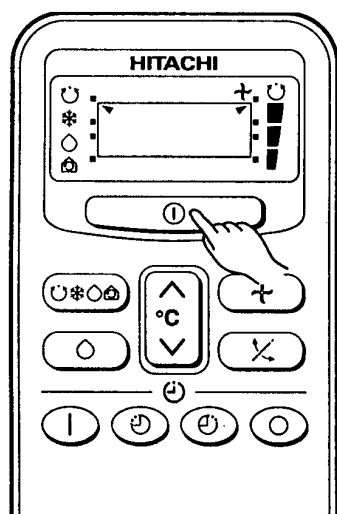
You can use the remote controller by fixing it on the wall with the accessory parts.

- Before fixing it, make sure the unit can be controlled by the remote controller at the fixing position.



## AUTOMATIC OPERATION

The device will automatically determine the operation mode either COOL or DEHUMIDIFY, depending on the initial room temperature.



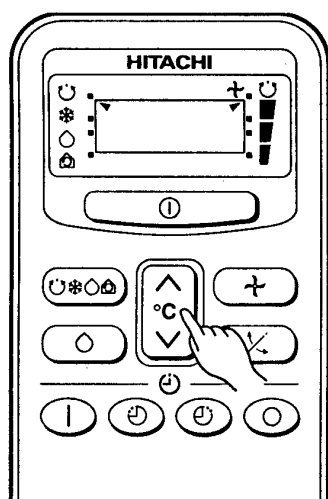
### 1 Press the (PRT/STOP) button

- Operation start with a “beep” sound.

### **STOP** Press the (P/STOP) button

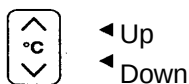
- Please make sure you have been selected the  (Automatic) mode by use the  (FUNCTION) Button.
- The fan speed selector does not work at this operation.

You can slightly adjust the room temperature.



### 1 Press the (TEMPERATURE) button

- Temperature setting change by 1 for each 1 time press.



- You can raise or lower the temperature setting by a maximum of 3 .
- The display does not indicate the preset temperature in the Automatic mode. Device will receive the setting by a “beep”.

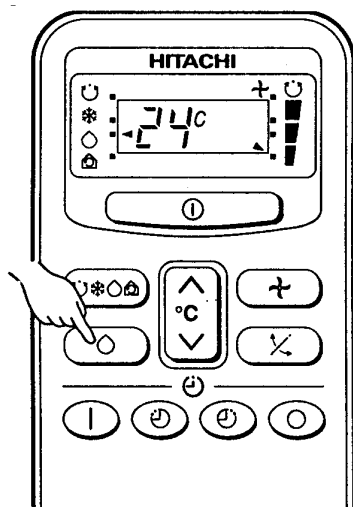
## CONDITION OF AUTOMATIC OPERATION

The selected mode of operation will not change during the operation even though the room temperature change.

| INITIAL ROOM TEMPERATURE (APPROX.) |                                                                                     | FUNCTION   | TEMPERATURE SETTING | FAN SPEED                                                  |
|------------------------------------|-------------------------------------------------------------------------------------|------------|---------------------|------------------------------------------------------------|
| Over 25                            |  | COOL       | 25                  | HIGH at start, LOW after the preset temperature is reached |
| 16~25                              |  | DEHUMIDIFY | 23                  | LOW                                                        |
| Under 16                           |  |            |                     | OFF                                                        |

## ENERGY SAVE DEHUMIDIFYING OPERATION

Use the device for dehumidifying when the room temperature is over 16 .



**1** Press the  HUMIDIFY) button

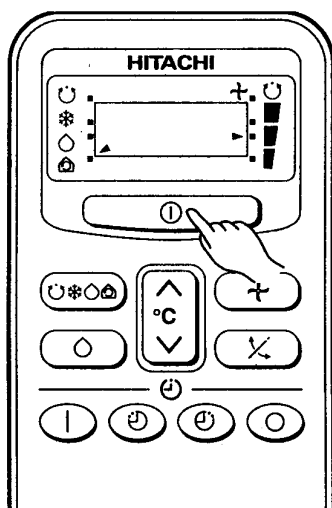
- Operation start with a “beep” sound.

**STOP** Press the  T/STOP) button

- when you want to change the operation mode, please use the  (FUNCTION) Button.
- Set the desired temperature is available.
- You also can use the function mode button to select this operation.

## CIRCULATION OPERATION

when the room temperature is higher than the thermostat set temperature, only the fan operates to circulate the upper hotter air downwards. This makes temperature distribution of the room evenly at heating.



**1** Press the  RT/STOP) button

- Operation start with a single received sound “beep”.
- **STOP** is the  T/STOP) button
- Make sure you have been select the  (circulate) mode by using the  (FUNCTION) Button.
- You can select the fan speed and desired temperature.
- This operation is useful in winter when using other heating equipment.





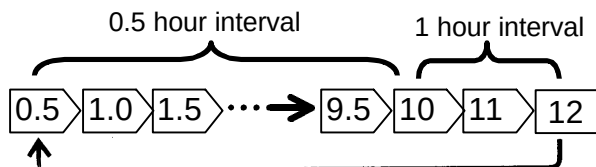
## TIMER RESERVATION

ON Timer and OFF Timer are available.

### OFF Timer Reservation

#### 1 OFF TIMER setting

- Select the OFF TIMER by pressing the  Button.
- Setting time will change according to the below sequence when you press the button.

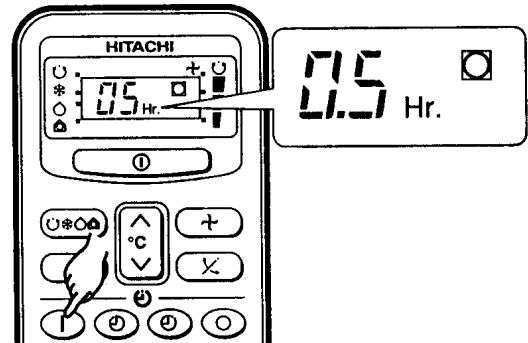
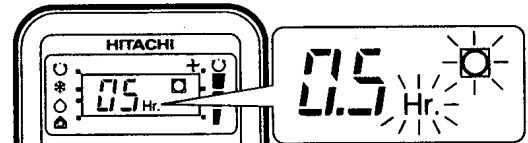


- The value change more quickly if you keep pressing the button.

#### 2 Press the button

- OFF TIMER reserved with a signal received sound "beep".
- The  Mark starts lighting instead of blinding.

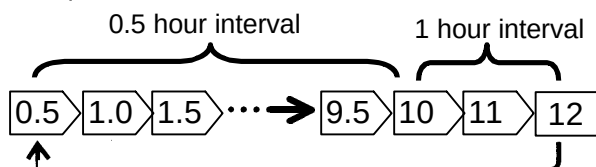
Operation stop at setting time



### ON Timer Reservation

#### 1 ON TIMER setting

- Select the ON TIMER by pressing the  Button.
- At the beginning of setting, time 6 hours was set.
- Setting time will change according to the below sequence.

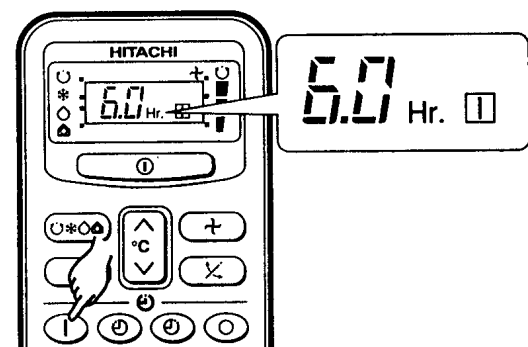
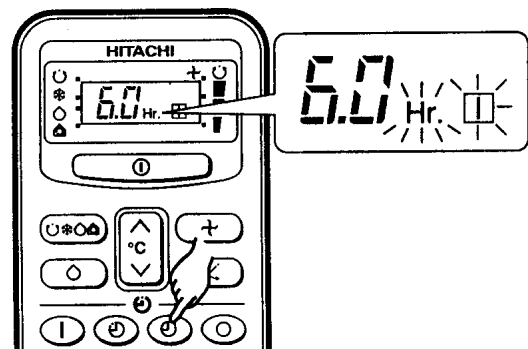


- The value change more quickly if you keep pressing the button.

#### 2 Press the button

- ON TIMER reserved with a signal received sound "beep".
- The  Mark starts lighting instead of blinding.

Operation will start for setting temperature at setting time (The starting time may different depend on the room temperature and set temperature).



### CANCELLATION OF Timer Reservation

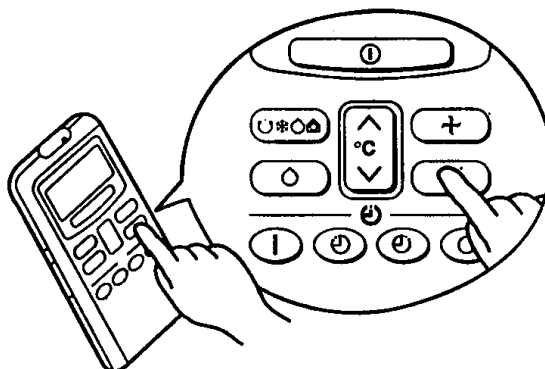
#### 1 Press the button

- As the time settings are stored in remote controller memory, you only have to press the  (Reserve) button in order to use the same setting next time.

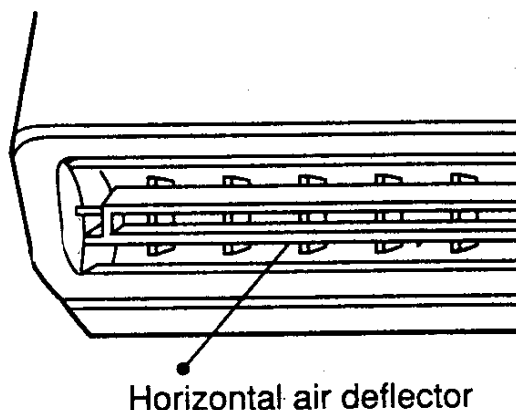
## ADJUSTING THE AIR DEFLECTOR

- 1 Adjustment of the conditioned air in the upward and downward directions.

According to Dehumidifying or Cooling operation, the horizontal air deflector is automatically set to the proper angle suitable for each operation. The deflector can be swung up and down and also set to the desired angle using the (AUTO SWING) button. (If the angle of the deflector is changed, it will not return to the auto-set position after operations start unless the operation mode is switched).

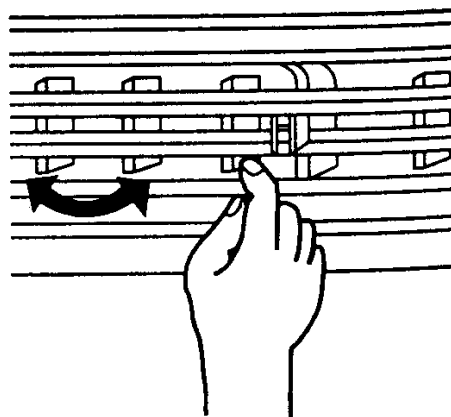


- If the (AUTO SWING) button is pressed once, the horizontal air deflector swings up and down. If the button is pressed again, the deflector stops in its current position. Several seconds (about 12 seconds) may be required before the deflector starts to move.
- Swinging the air deflector is effective to reduce unevenness of the temperature in the room.
- In Cooling operation, do not keep the horizontal air deflector swinging for a long time. Some dew may form on the horizontal air deflector and some dew drops may fall from it.
- When the (AUTO SWING) button is pressed while the operation is stopped, the horizontal air deflector moves and stops at the position where the air outlet closes.



- 2 Adjustment of the conditioned air to the left and right.

Hold the vertical air deflector as shown in the figure and adjust the conditioned air to the left and right.



## HOW TO EXCHANGE THE BATTERIES IN THE REMOTE CONTROLLER

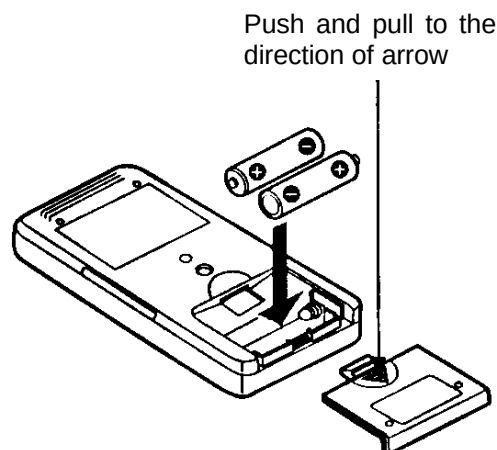
- 1 Remove the cover as shown in the figure and take out the old batteries.



- 2 Install the new batteries. The direction of the batteries should match the marks in the case.

### CAUTION

1. Do not use new and old batteries, or different kinds of batteries together.
2. Take out the batteries when you do not use the remote controller for 2 or 3 months.
3. The batteries must be of the LR03 type.



## TEMPORARY SWITCH WITH AUTO RESTART

Use the temporary switch with auto restart when operation can not be done with the remote controller.

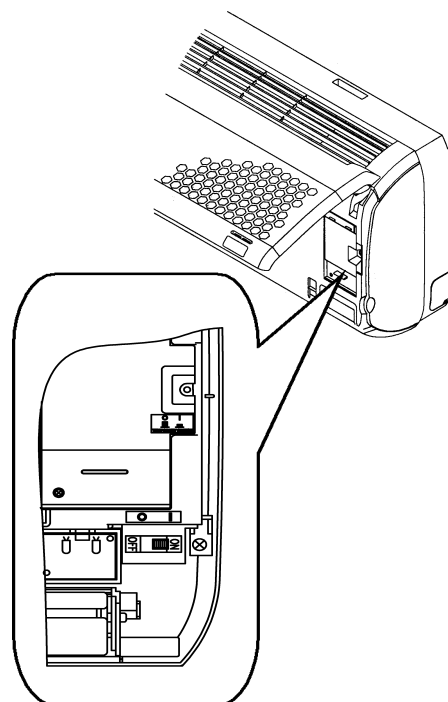
1. By pressing the temporary switch with auto restart, the operation is done in previously set operation mode.

When the operation is done using the temporary switch with auto restart after the power source is turned off and is turned on again, the operation is done in automatic mode.

2. When the operation is stopped or when the operation is done with the remote controller again, press the temporary switch once again.
3. Because the temporary switch has the function of auto restart, the power will be supplied again after power stoppage.

### NOTE

Although the Temporary switch with auto restart has been at "ON" position, you can also use the remote controller to make further setting.



## POWER SWITCH

When you do not use the room air conditioner, set the power switch to "O"(OFF).

## MAINTENANCE



### CAUTION

Before the cleaning, stop operation and disconnect the power supply.

Clean the air filter, as it removes dust inside the room. It should be washed when Air conditioner be used more than one month. In case the air filter is full of dust, the air flow will decrease and the cooling capacity will be reduced. Further, noise may occur. Be sure to clean the filter following the procedure below.

### Procedure

Remove the filter.

1

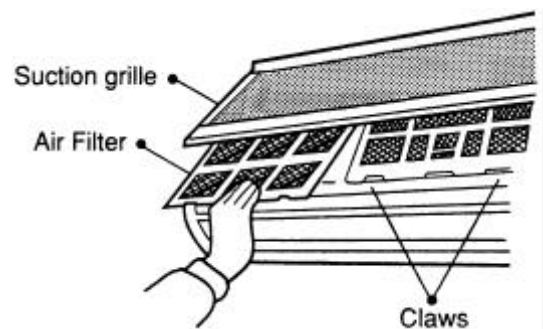
- Hold both sides of the suction grille and lift it up to open it.(Until the fixed position)
- Slightly lift the filter and release the claws(2 locations)at the lower part of the front cover and remove the filter from the lower side.



2

Remove dust from the filter using a vacuum-cleaner.

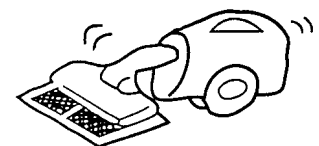
If there is too much dust, use neutral detergent.  
After using neutral detergent, wash with clean water and dry in the shade.



3

Install the filters.

Slightly lift the suction grille and close as original state.

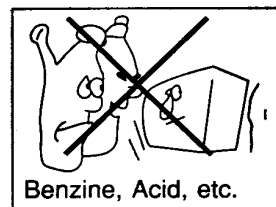
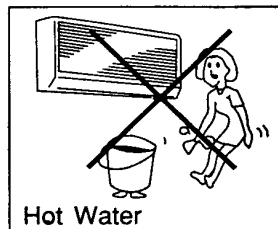


### CAUTION

- Do not wash with hot water at more than 40 . The filter may shrink.
- When washing it, shake off moisture completely and dry it in the shade; do not expose it directly to the sun. The filter may shrink.
- Do not operate the air conditioner with the filter removed. Dust may enter the air conditioner and cause trouble.

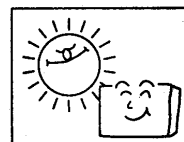
## CLEANING OF FRONT COVER, ETC.

- 1.The front cover easily becomes dirty. Wipe it with a soft dry cloth.
- 2.When it is excessively dirty, wipe with a soft cloth soaked in lukewarm water or neutral detergent. Then wipe thoroughly with a soft dry cloth.
- 3.Never use hot water (above 50 °C), benzine, gasoline, acid, thinner or a brush, because they will damage the plastic surface and the coating.



## MAINTENANCE AT BEGINNING OF LONG OFF PERIOD

- 1.Running the unit setting the operation mode to  $\oplus$ AN) and the fan speed to HI for about half a day on a fine day, and dry the whole of the unit.



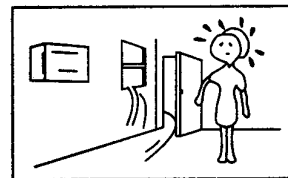
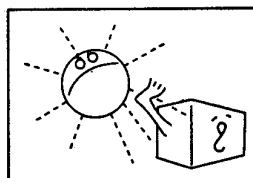
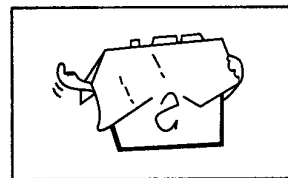
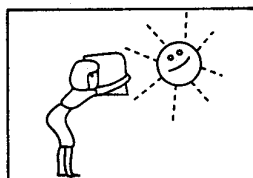
## WHEN ASKING FOR SERVICE, CHECK THE FOLLOWING POINTS.

### WHEN IT DOES NOT OPERATE

- 1.Is the fuse all right?
- 2.Is the voltage extremely high or low?
- 3.Is the power switch ON ?

### WHEN IT DOES NOT COOL WELL

- 1.Is the air filter blocked with dust?
- 2.Does sunlight fall directly on the outdoor unit?
- 3.Is the air flow of the outdoor unit obstructed?
- 4.Are the doors or windows opened, or is there any source of heat in the room?
- 5.Is the set temperature suitable?



### Notes

- In quiet operation or stopping the running, the following phenomena may occasionally occur, but they are not abnormal for the operation.
  - (1) Slight flowing noise of refrigerant in the refrigerating cycle.
  - (2) Slight rubbing noise from the fan casing which is cooled and then gradually warmed as operation stops.
- The odor will possibly be emitted from the room air conditioner because the various odor, emitted by smoke, foodstuffs, cosmetics and so on, sticks to it. So please clean the air filter and the evaporator at the beginning of the season to reduce the odor.

This product conforms with the protection requirements of council directives 89/336/EEC relating to electromagnetic compatibility.

Please note:

On switching on the equipment, particularly when the room light is dimmed, a slight brightness fluctuation may occur. This is of no consequence.

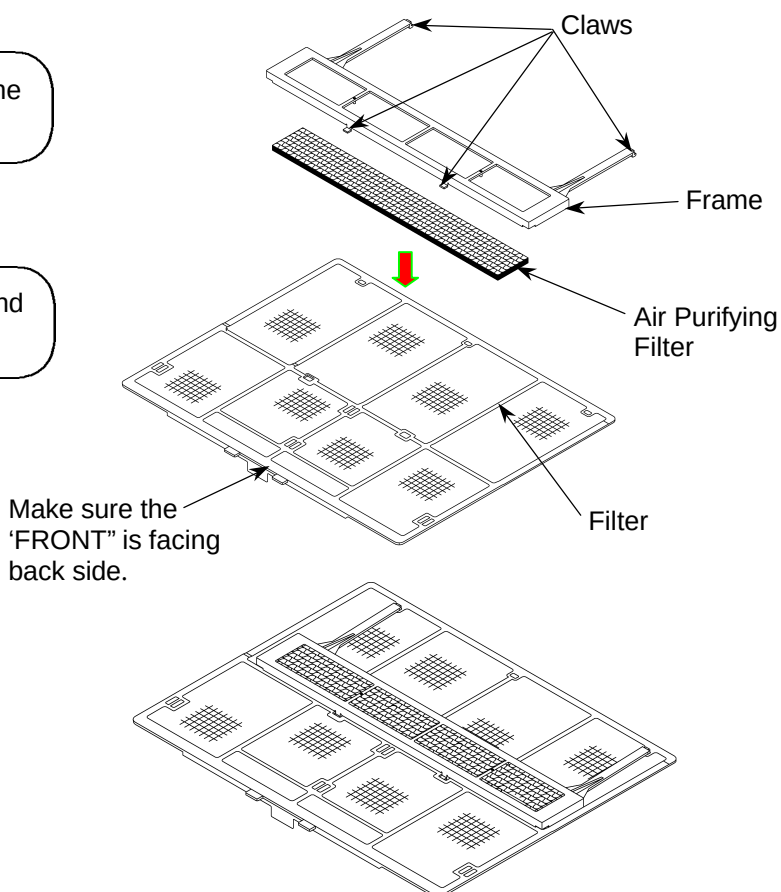
The conditions of the local Power Supply Companies are to be observed.

## HOW TO EXCHANGE THE AIR PURIFYING FILTER

- 1 Open the frame by releasing the claws from the filter holes.



- 2 Place the new air purifying filter and attach the frame back to the filter.

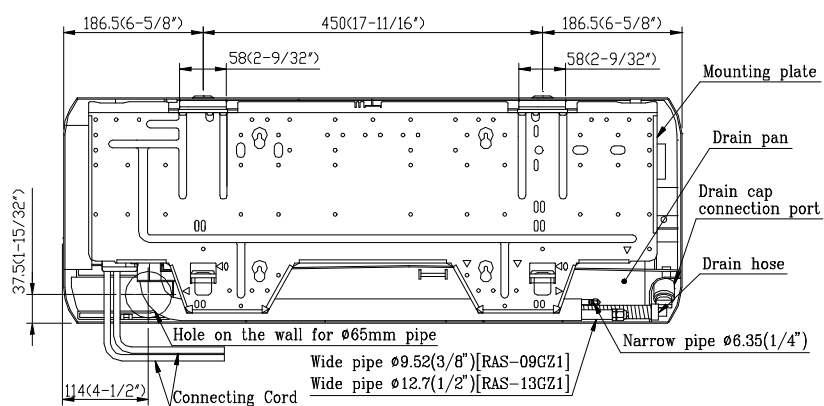


### NOTE:

- This model has an air purifying filter. The cooling capacity is slightly weakened and the cooling speed becomes slower when the air purifying filter is used. So, set the fan speed to "HIGH" when using it in this condition.
- Recommended to replace the air purifying filter after every 3 months for normal usage. Type number for this air purifying filter is <SPX-CFH5>. Please use this number for ordering when you want to renew it.

MODEL RAS-09GZ1  
RAS-13GZ1

Wireless remote controller



Technical drawing of a pipe rack showing front and back views with dimensions.

**FRONT VIEW (VIEWED FROM Ⓟ):**

- Overall width:  $134(5-9/32")$
- Inner width:  $60(2-3/8")$
- Overall height:  $47(1-27/32")$

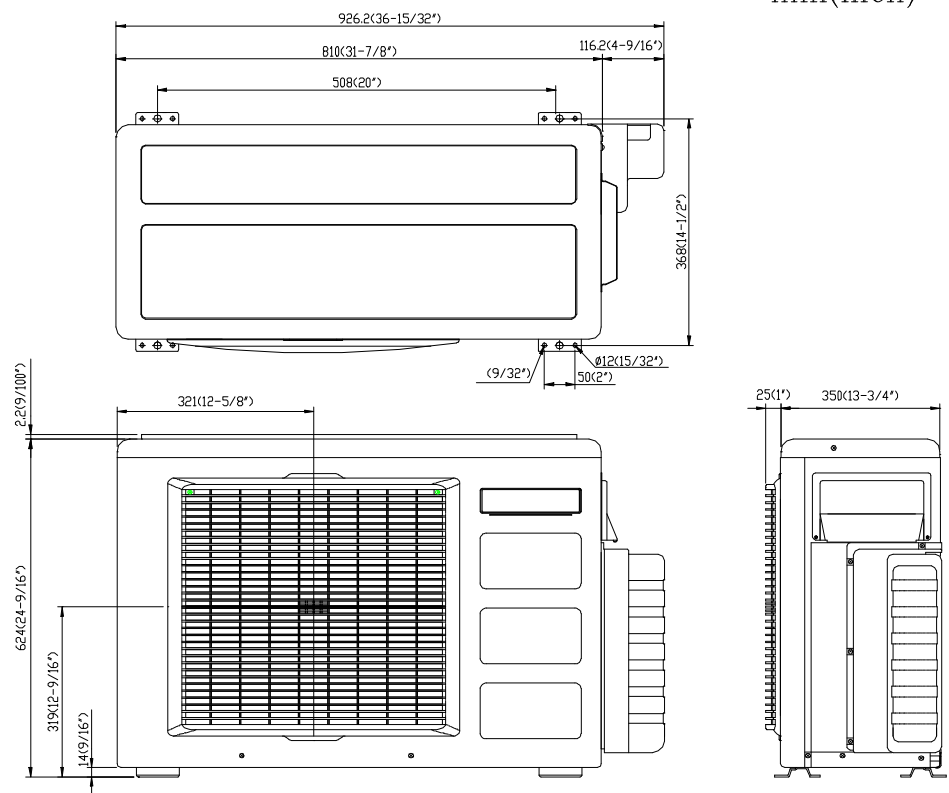
**BACK VIEW (VIEWED FROM BACK (pipe lead-out)):**

- Overall width:  $400(15-3/4")$
- Inner width:  $290(11-3/8")$
- Overall height:  $360(14-3/16")$

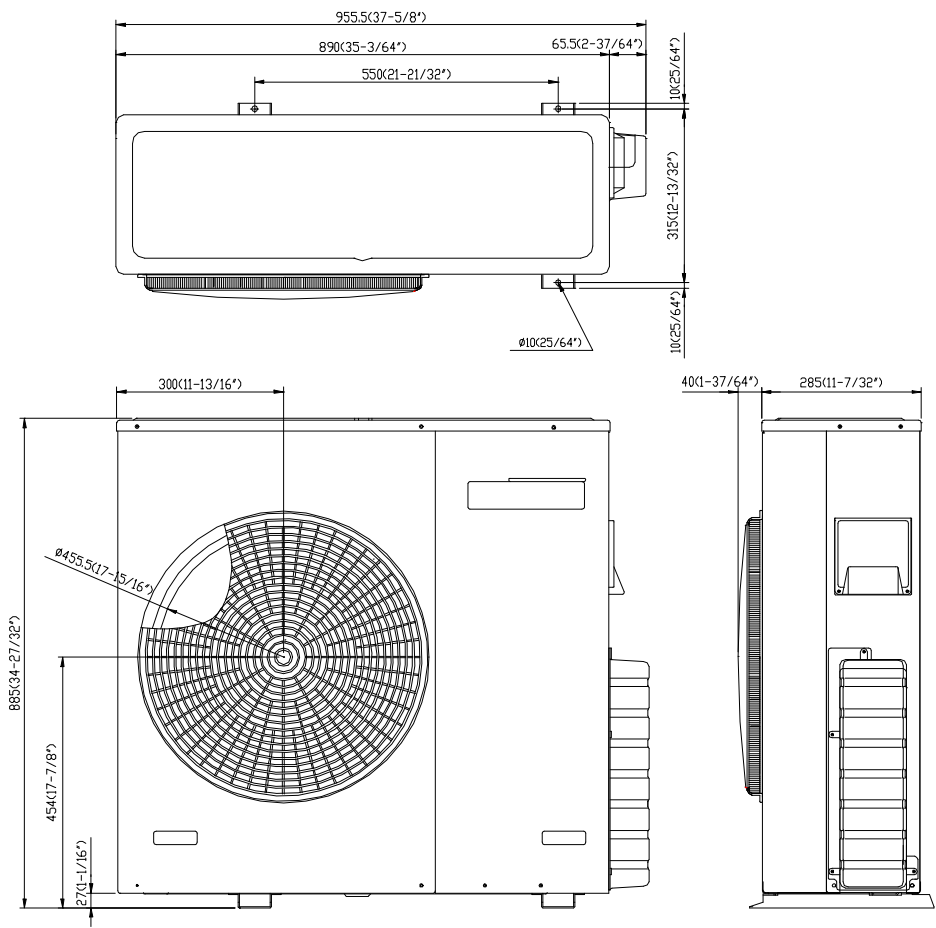
CONSTRUTION AND DIMENSIONAL DIAGRAM (Outdoor Unit )

MODEL RAM-20GW1

mm(inch)



MODEL RAM-30GW1





## MAIN PARTS COMPONENT

MODEL RAS-09GZ1 x 3 / RAM-20GW1

### THERMOSTAT

Thermostat Specifications

| MODEL              |               |     | RAS-09GZ1 x 3 |
|--------------------|---------------|-----|---------------|
| THERMOSTAT MODEL   |               |     | IC            |
| TEMPERATURE<br>( ) | INDICATION 16 | ON  | 17.0 (62.6)   |
|                    |               | OFF | 16.3 (61.3)   |
|                    | INDICATION 24 | ON  | 25.0 (77.0)   |
|                    |               | OFF | 24.3 (75.7)   |
|                    | INDICATION 32 | ON  | 33.0 (91.4)   |
|                    |               | OFF | 32.3 (90.4)   |

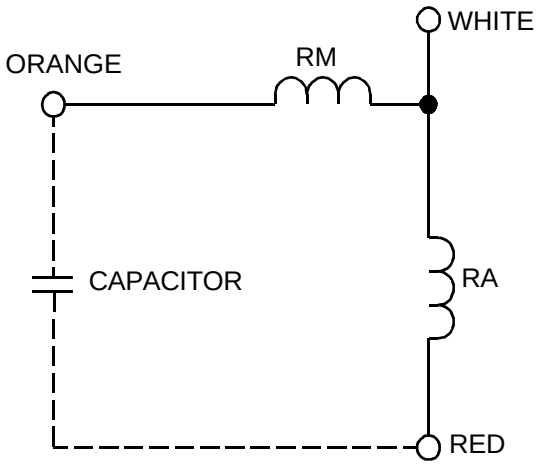
### FAN MOTOR

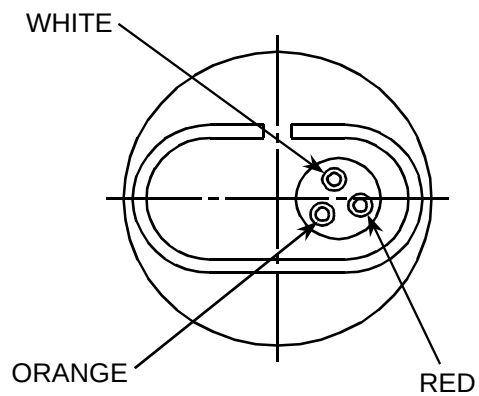
Fan Motor Specifications

| MODEL                   |              | RAS-09GZ1                               | RAM-20GW1            |
|-------------------------|--------------|-----------------------------------------|----------------------|
| PHASE                   |              | SINGLE                                  |                      |
| RATED VOLTAGE           |              | 220 - 230V                              |                      |
| RATED FREQUENCY         |              | 50Hz                                    |                      |
| OUT PUT                 |              | 8W                                      | 40W                  |
| POLE NUMBER             |              | 4                                       | 4                    |
| CONNECTION              |              |                                         |                      |
| RESISTANCE<br>VALUE (Ω) | 20<br>(68 )  | RM=605 RA2=57<br>RA1=77 RA3=249         | RA=146.2<br>RM=92.1  |
|                         | 75<br>(167 ) | RM=735.5 RA2=69.3<br>RA1=93.6 RA3=302.7 | RA=177.7<br>RM=112.0 |

## COMPRESSOR

### Compressor Motor Specifications

|                      |                                                                                     |                     |
|----------------------|-------------------------------------------------------------------------------------|---------------------|
| MODEL                | RAM-20GW1                                                                           |                     |
| COMPRESSOR MODEL     | GK516S1Z x 2                                                                        |                     |
| PHASE                | SINGLE                                                                              |                     |
| RATED VOLTAGE        | 220 - 240V                                                                          |                     |
| RATED FREQUENCY      | 50Hz                                                                                |                     |
| LOCKED ROTOR CURRENT | 22A                                                                                 |                     |
| POLE NUMBER          | 2                                                                                   |                     |
| CONNECTION           |  |                     |
| RESISTANCE VALUE(Ω)  | 20<br>(68 )                                                                         | RM=3.51<br>RA1=2.77 |
|                      | 75<br>(167 )                                                                        | RM=4.27<br>RA1=3.37 |



MODEL RAS-13GZ1 / RAM-30GW1  
RAS-09GZ1 x 2

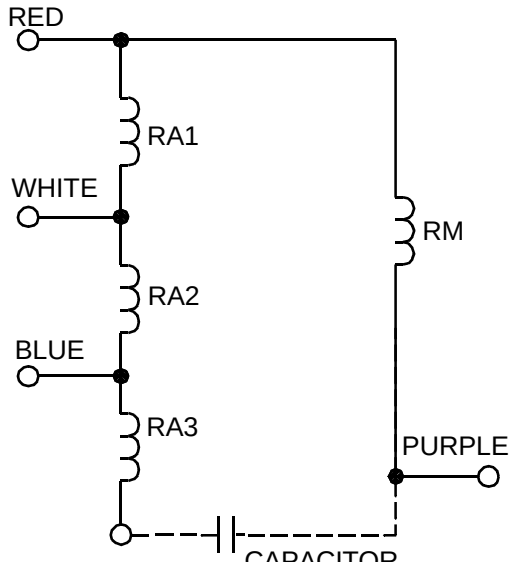
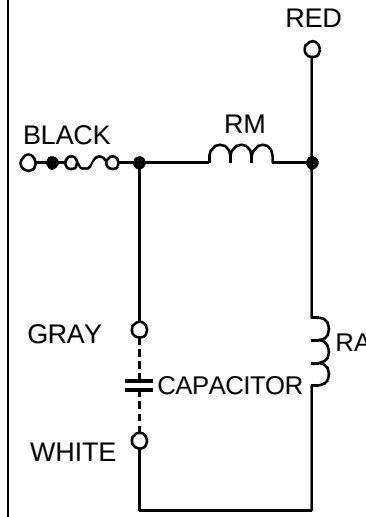
## THERMOSTAT

### Thermostat Specifications

| MODEL              |               |     | RAS-09GZ1   | RAS-13GZ1 |
|--------------------|---------------|-----|-------------|-----------|
| THERMOSTAT MODEL   |               |     | IC          |           |
| TEMPERATURE<br>( ) | INDICATION 16 | ON  | 17.0 (62.6) |           |
|                    |               | OFF | 16.3 (61.3) |           |
|                    | INDICATION 24 | ON  | 25.0 (77.0) |           |
|                    |               | OFF | 24.3 (75.7) |           |
|                    | INDICATION 32 | ON  | 33.0 (91.4) |           |
|                    |               | OFF | 32.3 (90.4) |           |

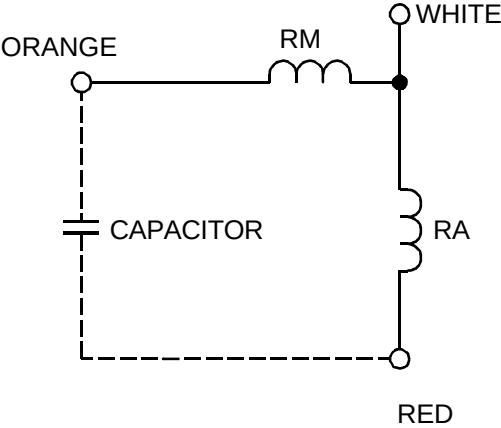
## FAN MOTOR

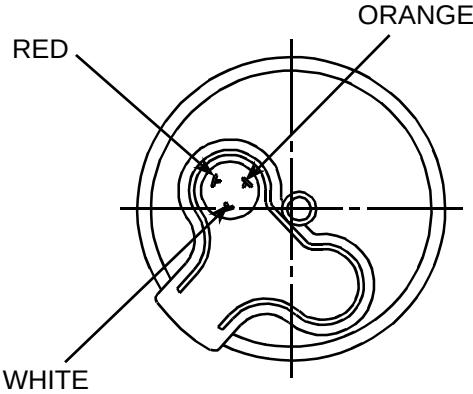
### Fan Motor Specifications

| MODEL                   | RAS-13GZ1                                                                            |                                         | RAS-09GZ1 x 2                           | RAM-30GW1                                                                             |
|-------------------------|--------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------|
| PHASE                   | SINGLE                                                                               |                                         |                                         |                                                                                       |
| RATED VOLTAGE           | 220 - 230V                                                                           |                                         |                                         |                                                                                       |
| RATED FREQUENCY         | 50Hz                                                                                 |                                         |                                         |                                                                                       |
| OUT PUT                 | 10W                                                                                  |                                         | 8W                                      | 75W                                                                                   |
| POLE NUMBER             | 4                                                                                    |                                         | 4                                       | 4                                                                                     |
| CONNECTION              |  |                                         |                                         |  |
| RESISTANCE<br>VALUE (Ω) | 20<br>(68 )                                                                          | RM=605 RA2=98<br>RA1=112 RA3=306        | RM=605 RA2=57<br>RA1=77 RA3=249         | RA=83.3<br>RM=53.1                                                                    |
|                         | 75<br>(167 )                                                                         | RM=735.5 RA2=119.1<br>RA1=136.2 RA3=372 | RM=735.5 RA2=69.3<br>RA1=93.6 RA3=302.7 | RA=101.3<br>RM=64.6                                                                   |

COMPRESSOR

Compressor Motor Specifications

|                      |              |                                                                                     |                       |
|----------------------|--------------|-------------------------------------------------------------------------------------|-----------------------|
| MODEL                |              | RAM-30GW1                                                                           |                       |
| COMPRESSOR MODEL     |              | 4KS210DAA                                                                           | 4KS340DAA             |
| PHASE                |              | SINGLE                                                                              |                       |
| RATED VOLTAGE        |              | 220 - 240V                                                                          |                       |
| RATED FREQUENCY      |              | 50Hz                                                                                |                       |
| LOCKED ROTOR CURRENT |              | 26A                                                                                 | 41A                   |
| POLE NUMBER          |              | 2                                                                                   |                       |
| CONNECTION           |              |  |                       |
| RESISTANCE VALUE(Ω)  | 20<br>(68 )  | RM=2.159<br>RA1=3.527                                                               | RM=1.259<br>RA1=2.398 |
|                      | 75<br>(167 ) | RM=2.624<br>RA1=4.288                                                               | RM=1.531<br>RA1=2.915 |

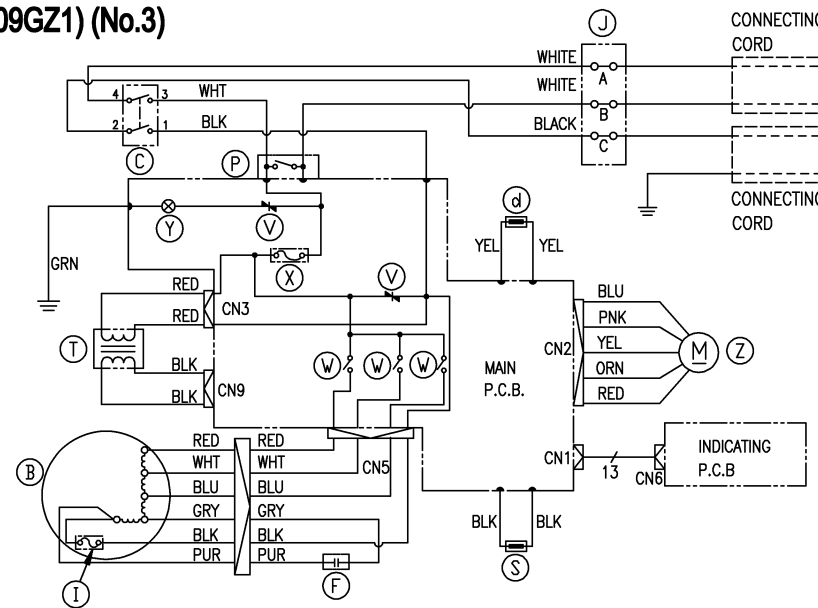


## WIRING DIAGRAM

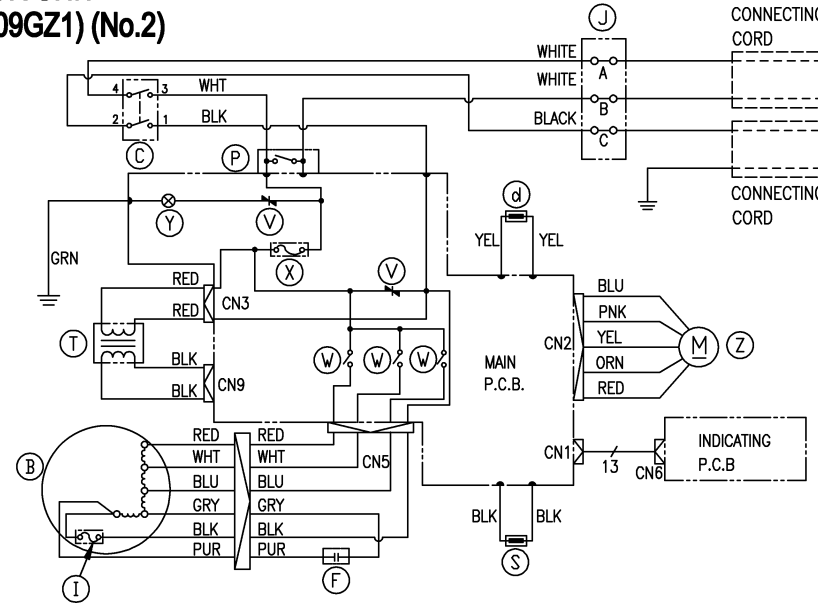
MODEL RAS-09GZ1×3 / RAM-20GW1

|                |                     |     |        |
|----------------|---------------------|-----|--------|
| Ⓐ              | COMPRESSOR          | BLU | BLUE   |
| Ⓑ              | FAN MOTOR           | YEL | YELLOW |
| Ⓒ              | POWER SWITCH        | BRN | BROWN  |
| Ⓕ              | 1μF CAPACITOR       | WHT | WHITE  |
| Ⓖ              | 40μF CAPACITOR      | GRY | GRAY   |
| Ⓗ              | 2.3μF CAPACITOR     | ORN | ORANGE |
| Ⓘ              | FAN MOTOR PROTECTOR | GRN | GREEN  |
| Ⓙ              | TERMINAL BOARD      | RED | RED    |
| Ⓚ              | POWER CORD          | BLK | BLACK  |
| Ⓟ              | POWER RELAY         | PNK | PINK   |
| Ⓡ              | 2.3μF CAPACITOR     | PUR | PURPLE |
| Ⓢ              | THERMISTOR          |     |        |
| Ⓣ              | TRANSFORMER         |     |        |
| Ⓤ <sub>p</sub> | OVERLOAD RELAY      |     |        |
| Ⓥ              | VARISTOR            |     |        |
| Ⓦ              | FAN RELAY           |     |        |
| Ⓧ              | FUSE                |     |        |
| Ⓨ              | SURGE ABSORBER      |     |        |
| Ⓩ              | AUTO SWEEP MOTOR    |     |        |
| ⓓ              | THERMOSTAT          |     |        |
| Ⓝ              | 2-WAY VALVE COIL    |     |        |

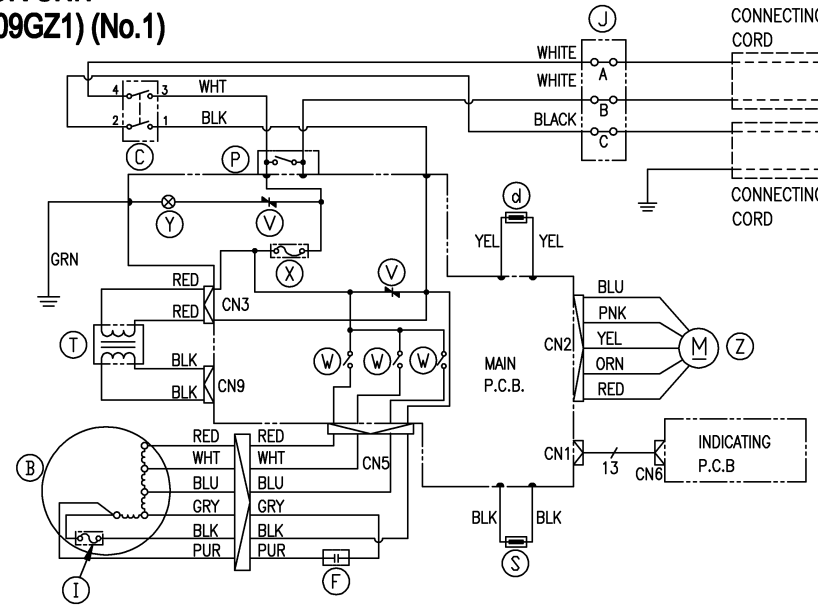
**INDOOR UNIT  
(RAS-09GZ1) (No.3)**



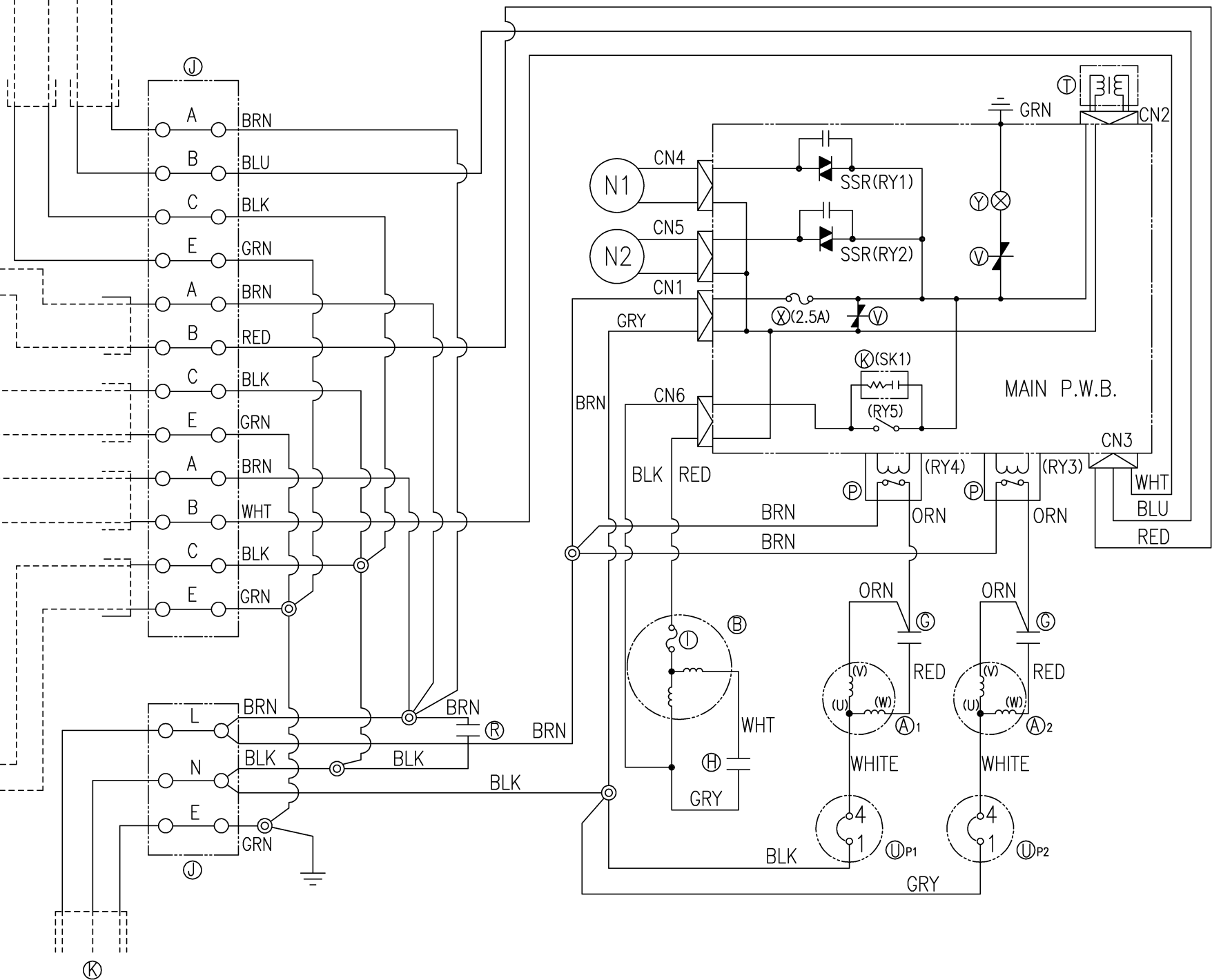
**INDOOR UNIT  
(RAS-09GZ1) (No.2)**



**INDOOR UNIT  
(RAS-09GZ1) (No.1)**



**OUTDOOR UNIT  
(RAM-20GW1)**

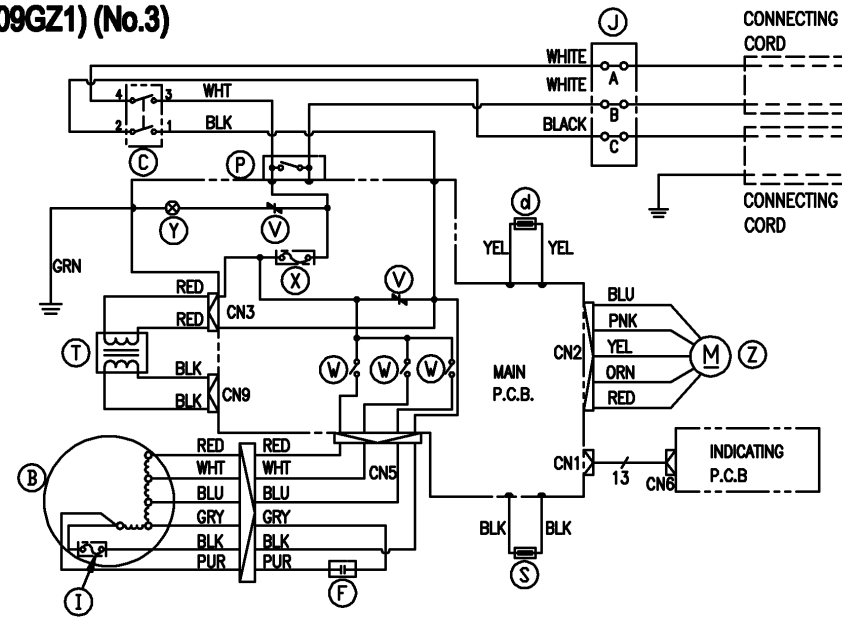


## WIRING DIAGRAM

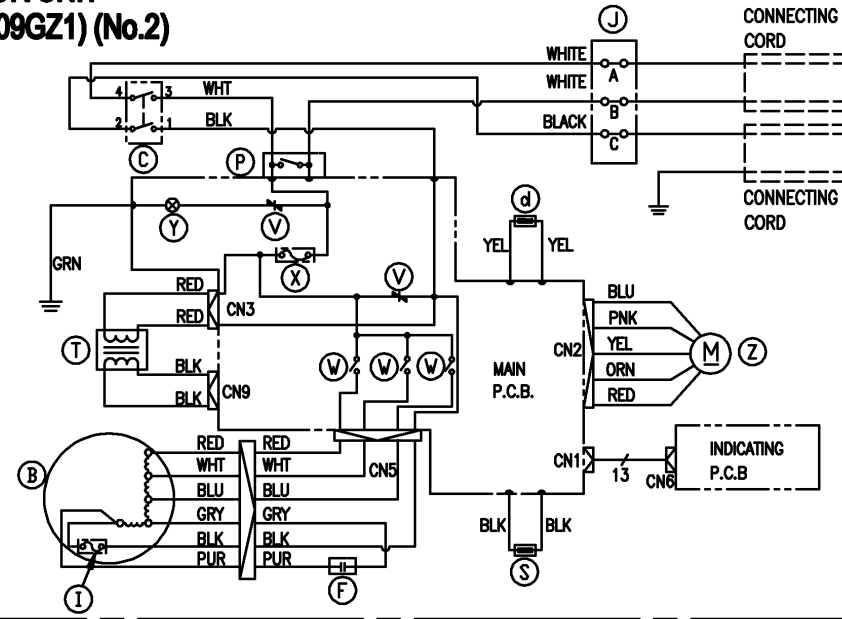
MODEL RAS-13GZ1  
RAS-09GZ1x 2 / RAM-30GW1

|                  |                         |            |
|------------------|-------------------------|------------|
| (A) <sub>1</sub> | COMPRESSOR( 4KS210DAA ) | BLU BLUE   |
| (A) <sub>2</sub> | COMPRESSOR( 4KS340DAA ) | YEL YELLOW |
| (B)              | FAN MOTOR               | BRN BROWN  |
| (C)              | POWER SWITCH            | WHT WHITE  |
| (F)              | 1 $\mu$ F CAPACITOR     | GRY GRAY   |
| (G) <sub>1</sub> | 35 $\mu$ F CAPACITOR    | ORN ORANGE |
| (G) <sub>2</sub> | 60 $\mu$ F CAPACITOR    | GRN GREEN  |
| (H)              | 4 $\mu$ F CAPACITOR     | RED RED    |
| (I)              | FAN MOTOR PROTECTOR     | BLK BLACK  |
| (J)              | TERMINAL BOARD          | PNK PINK   |
| (K)              | POWER CORD              | PUR PURPLE |
| (P)              | POWER RELAY             |            |
| (R)              | 2.3 $\mu$ F CAPACITOR   |            |
| (S)              | THERMISTOR              |            |
| (T)              | TRANSFORMER             |            |
| (V)              | VARISTOR                |            |
| (W)              | FAN RELAY               |            |
| (X)              | FUSE                    |            |
| (Y)              | SURGE ABSORBER          |            |
| (Z)              | AUTO SWEEP MOTOR        |            |
| (d)              | THERMOSTAT              |            |
| (N)              | 2-WAY VALVE COIL        |            |

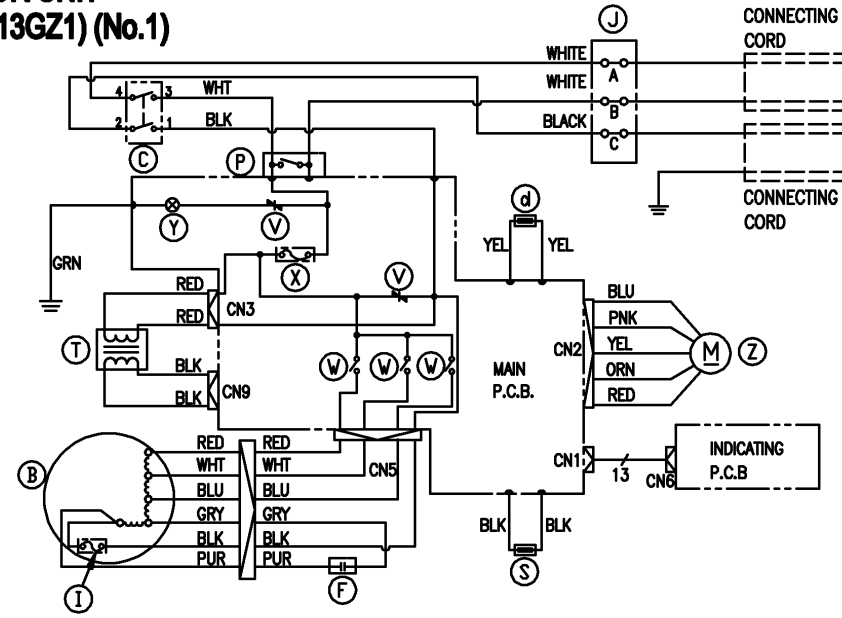
INDOOR UNIT  
(RAS-09GZ1) (No.3)



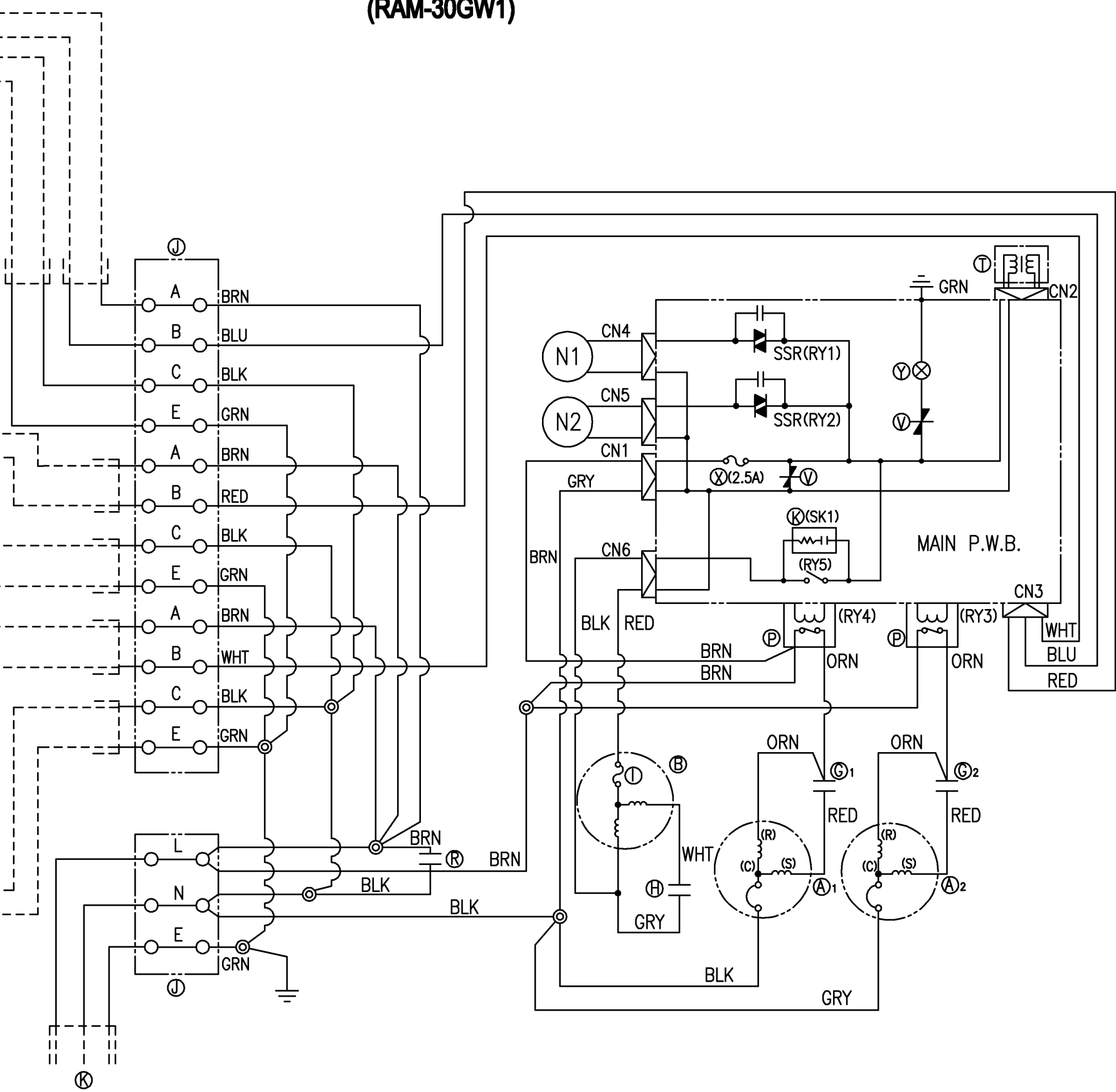
INDOOR UNIT  
(RAS-09GZ1) (No.2)



INDOOR UNIT  
(RAS-13GZ1) (No.1)

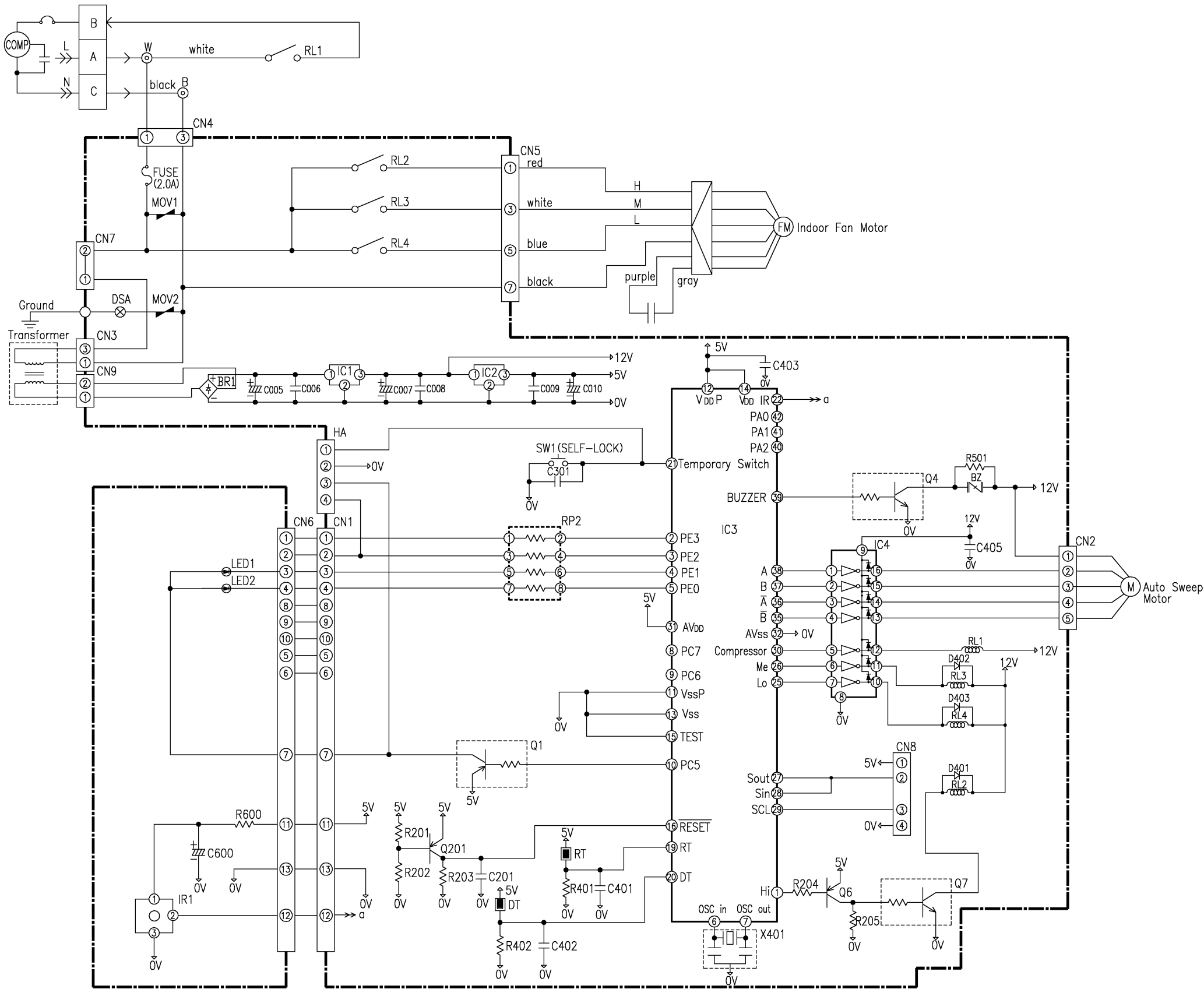


OUTDOOR UNIT  
(RAM-30GW1)





WIRING DIAGRAW OF THE PRINTED WIRING BOARD  
MODEL RAS-09GZ1, RAS-13GZ1



RESISTANCE(Ω)

| SYMBOL | RATING    | TYPE |
|--------|-----------|------|
| RP2    | 220Ω-5%   | H    |
| R201   | 10KΩ-1%   | R    |
| R202   | 61.9KΩ-1% | R    |
| R203   | 10KΩ-1%   | R    |
| R204   | 1.2K-5%   | R    |
| R205   | 10K-1%    | R    |
| R401   | 5.16KΩ-1% | R    |
| R402   | 5.16KΩ-1% | R    |
| R501   | 1.2KΩ-5%  | R    |
| R600   | 47Ω-5%    | H    |

| SYMBOL | MODEL          | TYPE |
|--------|----------------|------|
| FUSE   | 2.0A/250V      | H    |
| MOV1   | OS470NR        | H    |
| MOV2   | OS470NR        | H    |
| DSA    | DSA-362MA      | H    |
| BR1    | SIVB20         | H    |
| IC1    | 7812           | H    |
| IC2    | 7805           | H    |
| IC3    | CPU            | H    |
| IC4    | 2003           | H    |
| Q201   | 2SA1015        | R    |
| Q1     | DTA114TS(10KΩ) | R    |
| Q6     | 2SA1015        | R    |
| Q4,Q7  | DTC114TS(10KΩ) | R    |
| RL2    | ALD-112        | H    |
| RL3    | ALD-112        | H    |
| RL4    | ALD-112        | H    |
| X401   | 6M Hz          | R    |
| RT     | 5KΩ at 25      | H    |
| DT     | 5KΩ at 25      | H    |

CAPACITOR(F)

| SYMBOL | RATING      | TYPE |
|--------|-------------|------|
| C005   | ELE 2200i F | H    |
| C006   | 0.01i F     | R    |
| C007   | ELE 1000i F | H    |
| C008   | 0.1i F      | R    |
| C009   | ELE 470i F  | H    |
| C010   | 0.1i F      | R    |
| C201   | 0.1i F      | R    |
| C301   | 0.1i F      | R    |
| C401   | 2200PF      | R    |
| C402   | 2200PF      | R    |
| C403   | 0.1i F      | R    |
| C405   | 0.1i F      | R    |
| C600   | ELE 47i F   | H    |

DIODE

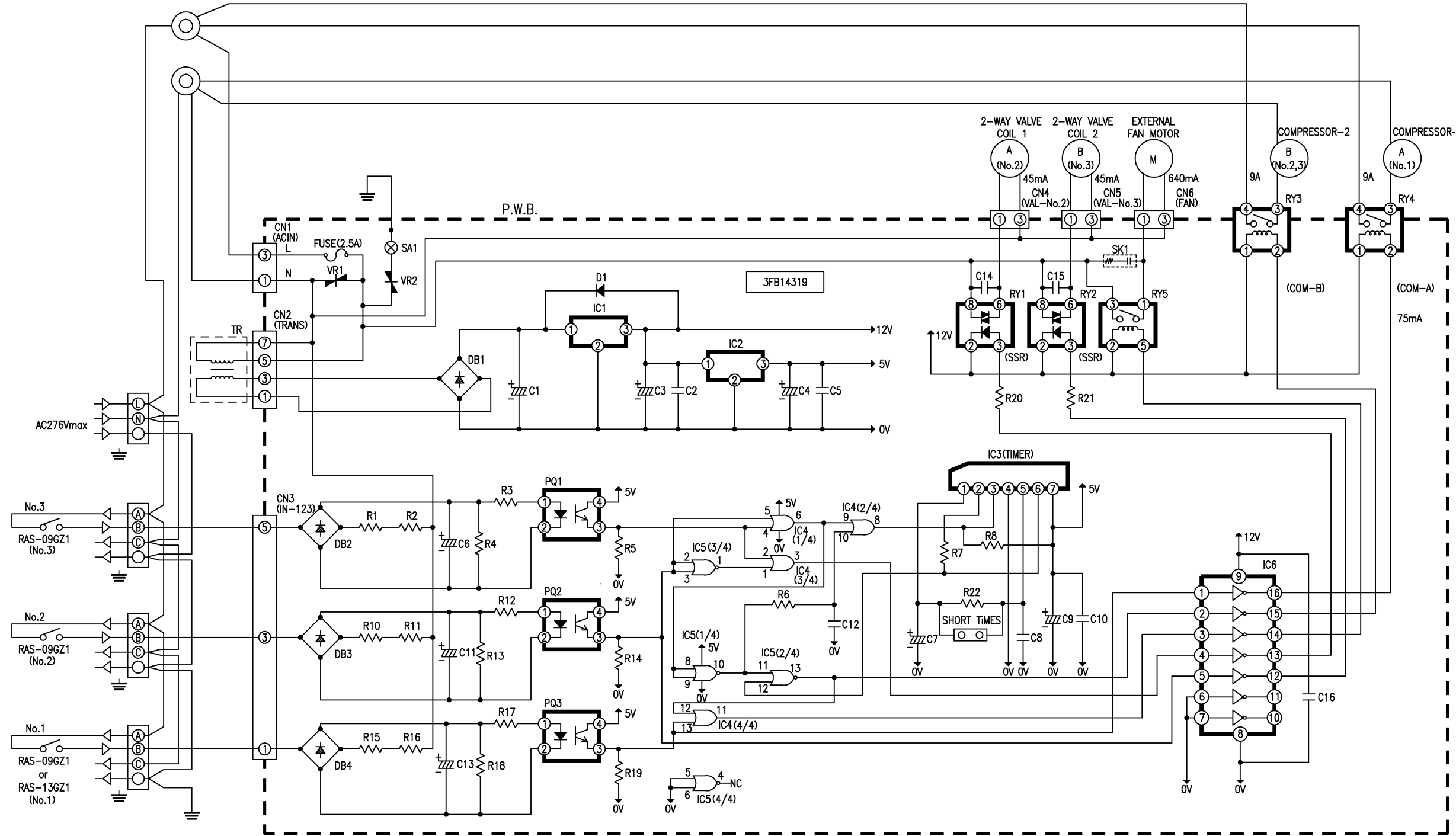
| SYMBOL | MODEL  |
|--------|--------|
| D401   | 1N4003 |
| D402   | 1N4003 |
| D403   | 1N4003 |

LED

| SYMBOL | MODEL           | COLOR  |
|--------|-----------------|--------|
| LED1   | L-44GD-LA6SF2.5 | GREEN  |
| LED2   | L-44YD-LA6SF2.5 | YELLOW |

H : (INSERT OF HAND)  
R : (RADIAL)

MODEL RAM-20GW1  
RAM-30GW1



J----5%  
RESISTANCE (Ω) F----1%

| SYMBOL | RATING      |
|--------|-------------|
| R1     | 24k J 2W    |
| R2     | 24k J 2W    |
| R3     | 560 J 1/4W  |
| R4     | 100K J 1/4W |
| R5     | 5.1k F 1/4W |
| R6     | 33k J 1/4W  |
| R7     | 5.6k J 1/4W |
| R8     | 5.6k J 1/4W |
| R10    | 24k J 2W    |
| R11    | 24k J 2W    |
| R12    | 560 J 1/4W  |
| R13    | 100k J 1/4W |
| R14    | 5.1k F 1/4W |
| R15    | 24k J 2W    |
| R16    | 24k J 2W    |
| R17    | 560 J 1/4W  |
| R18    | 100k J 1/4W |
| R19    | 5.1k F 1/4W |
| R20    | 620 J 1/2W  |
| R21    | 620 J 1/2W  |
| R22    | 150k F 1/4W |

D---ELECTROLYTIC CAPACITOR  
C---CERAMIC CAPACITOR  
F---FILM CAPACITOR

| CAPACITOR |               |      |  |
|-----------|---------------|------|--|
| SYMBOL    | RATING        | TYPE |  |
| C1        | 1000μ F 35V   | D    |  |
| C2        | 0.1μ F 25V    | C    |  |
| C3        | 100μ F 50V    | D    |  |
| C4        | 10μ F 16V     | D    |  |
| C5        | 0.1μ F 25V    | C    |  |
| C6        | 47μ F 16V     | D    |  |
| C7        | 10μ F 16V     | D    |  |
| C8        | 0.1μ F 50V    | F    |  |
| C9        | 10μ F 16V     | D    |  |
| C10       | 0.1μ F 25V    | C    |  |
| C11       | 47μ F 16V     | D    |  |
| C12       | 0.1μ F 50V    | F    |  |
| C13       | 47μ F 16V     | D    |  |
| C14       | 1000pF AC250V | C    |  |
| C15       | 1000pF AC250V | C    |  |
| C16       | 0.1μ F 25V    | C    |  |

| IC     |        |
|--------|--------|
| SYMBOL | MODEL  |
| IC1    | 7812   |
| IC2    | 7805   |
| IC3    | AN6780 |
| IC4    | 7432   |
| IC5    | 7402   |
| IC6    | 2003   |

| DIODE  |        |
|--------|--------|
| SYMBOL | MODEL  |
| D1     | IN4003 |

| VARISTOR |       |
|----------|-------|
| SYMBOL   | MODEL |
| VR1      | 470NR |
| VR2      | 470NR |

| DIODE BRIDGE |        |
|--------------|--------|
| SYMBOL       | MODEL  |
| DB1          | S1WB60 |
| DB2          | S1WB60 |
| DB3          | S1WB60 |
| DB4          | S1WB60 |

| RELAY  |                 |
|--------|-----------------|
| SYMBOL | MODEL           |
| RY1    | S26MD02V        |
| RY2    | S26MD02V        |
| RY3    | JM1aN-TMP-DC12V |
| RY4    | JM1aN-TMP-DC12V |
| RY5    | JS1-12V         |

| NOISE SUPPRESSOR |        |
|------------------|--------|
| SYMBOL           | MODEL  |
| SK1              | XE1201 |

| SURGE ABSORBER |            |
|----------------|------------|
| SYMBOL         | MODEL      |
| SA1            | S-ABSO-362 |

| PHOTO COUPLER |         |
|---------------|---------|
| SYMBOL        | MODEL   |
| PQ1           | PS-2501 |
| PQ2           | PS-2501 |
| PQ3           | PS-2501 |

| CONNECTOR |         |       |
|-----------|---------|-------|
| SYMBOL    | MODEL   | COLOR |
| CN1       | B2P3-VH | WHITE |
| CN2       | B4P7-VH | WHITE |
| CN3       | B3P5-VH | WHITE |
| CN4       | B2P3-VH | BLUE  |
| CN5       | B2P3-VH | RED   |
| CN6       | B2P3-VH | BLACK |

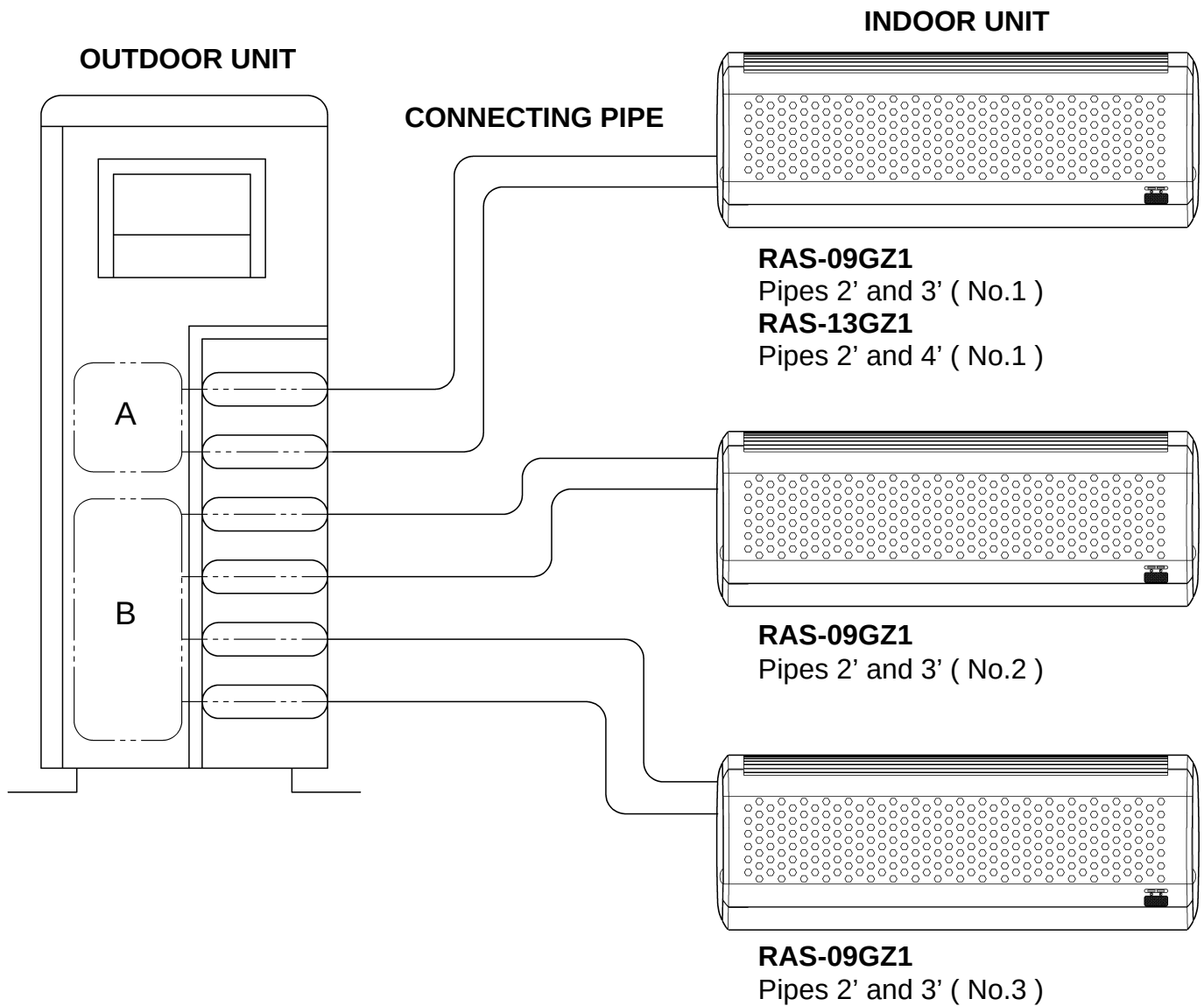
| FUSE   |           |
|--------|-----------|
| SYMBOL | RATING    |
| FUSE   | 2.5A/250V |

BASIC MODE

MODEL RAS-09GZ1 / RAS-13GZ1 / RAS-18GZ1 / RAS-25GZ1 / RAS-35GZ1 / RAS-09GZ1 x 2 / RAM-20GW1 / RAM-25GW1 / RAM-30GW1 ,

| Operation mode   |                                                         | Fan                                                                                                                                                                                                                                   | Cooling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sensor dehumidification | Automatic                                                                                                        |
|------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------|
| Control function |                                                         |                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |                                                                                                                  |
| 1                | Start/stop button, basic mode                           | <div><div>Start/Stop button</div><div>Operation lamp</div><div>Operation and stop are repeated</div><div>Independent of timer reservations</div></div>                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |                                                                                                                  |
| 2                | Timer operation                                         | OFF timer                                                                                                                                                                                                                             | <div><div>Start/Stop switch</div><div>Reservation</div><div>Cancel</div><div>Operation lamp</div><div>Timer lamp</div><div>Timer memory</div><div>(OFF timer during stop)</div><div>(Cancel)</div><div>(Change of the reservation time)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |                                                                                                                  |
|                  |                                                         | ON timer                                                                                                                                                                                                                              | <div><div>Start/Stop switch</div><div>Reservation</div><div>Cancel</div><div>Operation lamp</div><div>Timer lamp</div><div>Timer memory</div><div>(ON timer during operation)</div><div>(Cancel)</div><div>(Change of the reservation time)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                         |                                                                                                                  |
| 3                | Circulation mode                                        | Automatic                                                                                                                                                                                                                             | “HI”, ”MED” ,or “LO” operation is executed according to the thermostat signal.(Refer to “Thermostat operation”.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         | “HI”, ”MED” ,or “LO” operation is executed according to the thermostat signal.(Refer to “Thermostat operation”.) |
|                  |                                                         | HI                                                                                                                                                                                                                                    | Operation in “HI” mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Same as on the left.    | Same as on the left.                                                                                             |
|                  |                                                         | MED                                                                                                                                                                                                                                   | Operation in “MED” mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Same as on the left.    | Same as on the left.                                                                                             |
|                  |                                                         | LO                                                                                                                                                                                                                                    | Operation in “LO” mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Same as on the left.    | Same as on the left.                                                                                             |
| 4                | Thermostat operation<br>H HI<br>L MED<br>S LO           | Independent of the thermostat signal.                                                                                                                                                                                                 | <div><div>In case of “Automation” mode.</div><div><div>Set temperatuer 0</div><div>Start/ Stop switch</div><div>Fan relay H</div><div>Fan relay L</div><div>Fan relay S</div><div>Operation lamp</div><div>0.5sec</div><div>0.5sec</div></div><div><div>When the room temperature &lt; 21</div><div>Start/Stop</div><div>Power relay</div><div>FAN relay S</div><div>Operation lamp</div><div>3 min.</div><div>3 min.</div><div>0.5sec</div><div>0.5sec</div></div><div><div>When the room temperature 21</div><div>Set temperature</div><div>Start/ Stop switch</div><div>Fan relay S</div><div>Power relay</div><div>Operation lamp</div><div>Min. 3minutes</div><div>0.5sec</div><div>0.5sec</div></div><div><div>Notes: Room temperature : RT</div><div>Set temperature : ST</div><div>from Remote Controller</div><div>If RT ST+1</div><div>Then set temperature = ST</div><div>If RT &lt; ST+1</div><div>Then set temperature = RT-2</div></div></div> |                         |                                                                                                                  |
| 5                | Defrosting<br>( including automation fresh defrosting). | <div><div>Indoor heat exchanger thermistor temperature</div><div>T2</div><div>T1</div><div>Fan relay</div><div>Power relay</div><div>Operation lamp</div><div>5</div><div>10</div><div>10 minutes</div><div>Min. 3minutes</div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         | defrosting of each operation mode is executed.                                                                   |

**Multi piping condition when connecting 3 indoor units (3 rooms ) with 2 compressors.**



**RAM-20GW1**

Compressor A : Operates 1 indoor unit with 1 compressor ( Pipes 2' and 3' )

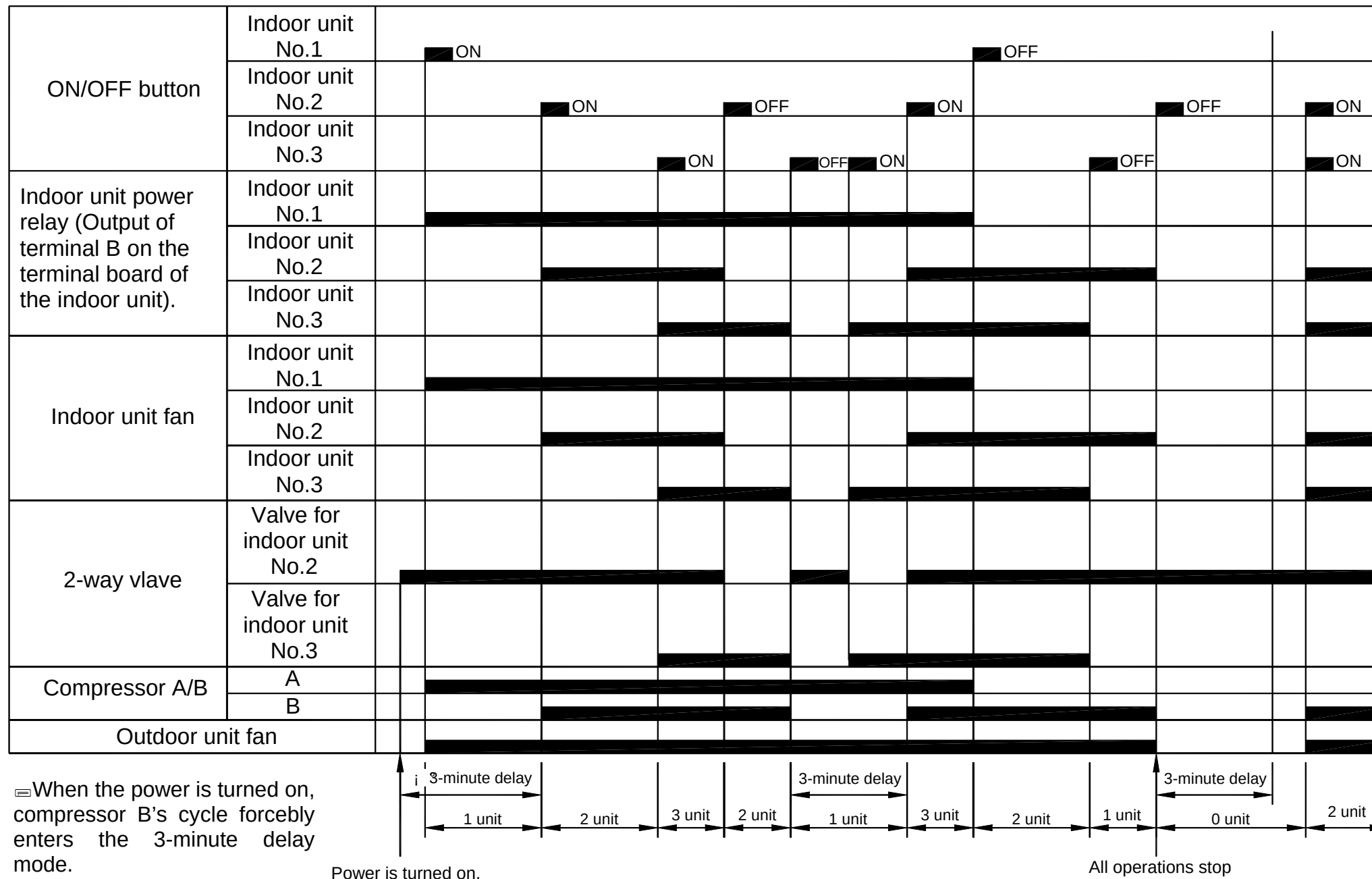
Compressor B : Operates 2 indoor unit with 1 compressor ( Pipes 2' and 3' )

**RAM-30GW1**

Compressor A : Operates 1 indoor unit with 1 compressor ( Pipes 2' and 4' )

Compressor B : Operates 2 indoor unit with 1 compressor ( Pipes 2' and 3' )

**Combination operation    Operation mode : Cooling**



- ⇒ When the power is turned on, compressor B's cycle forcibly enters the 3-minute delay mode.

Power is turned on.

All operations stop

### Open/close control of 2-way valve

|                  |                  |            |            |
|------------------|------------------|------------|------------|
| Indoor unit No.2 | Indoor unit No.3 | Valve No.2 | Valve No.3 |
| OFF              | OFF              | ON         | OFF        |
| ON               | OFF              | ON         | OFF        |
| OFF              | ON               | OFF        | ON         |
| ON               | ON               | ON         | ON         |

- As shown on the left, when both indoor units No.2 and No.3 are OFF, Valve No.2 is left open.
- When either of indoor unit No.2 or No.3 is ON, the valves corresponding to the indoor unit is opened.
- When both indoor units No.2 and No.3 are ON, both No.2 and No.3 valves are opened.

Note:

1. When the power is turned on, compressor B's cycle forcibly enters the 3-minute delay mode to perform initial setting for 2-way valve control.
2. Only compressor B's cycle is subject to the 3-minute delay. The 2-way valve for indoor unit No.2 is always opened during the 3-minute delay mode after compressor B has stopped. If the indoor unit is turned on during the 3-minute delay mode, the 2-way valve performs valve control to open the valve for indoor unit No.3 and close the valve for indoor unit No.2.
3. During the 3-minute delay mode after compressor B has stopped, even if indoor units No.2 and No.3, compressor B does not operate. The indoor units enter the ventilation (fan operation) mode.
4. Compressor A: Operates 1 indoor unit with 1 compressor (Pipes 2' and 3' for RAM-20GW1 ,Pipes 2' and 4' for RAM-30GW1 )  
Compressor B: Operates 2 indoor unit with 1 compressor (Pipes 2'and 3')

### Combination operation table (Cooling)

| Indoor unit No.1 | Indoor unit No.2 | Indoor unit No.3 | Compressor A | Compressor B | Valve No.2 | Valve No.3 | Outdoor unit fan |
|------------------|------------------|------------------|--------------|--------------|------------|------------|------------------|
| OFF              | OFF              | OFF              | OFF          | OFF          | ON         | OFF        | OFF              |
| OFF              | OFF              | ON               | OFF          | ON           | OFF        | ON         | ON               |
| OFF              | ON               | OFF              | OFF          | ON           | ON         | OFF        | ON               |
| OFF              | ON               | ON               | OFF          | ON           | ON         | ON         | ON               |
| ON               | OFF              | OFF              | ON           | OFF          | ON         | OFF        | ON               |
| ON               | OFF              | ON               | ON           | ON           | OFF        | ON         | ON               |
| ON               | ON               | OFF              | ON           | ON           | ON         | OFF        | ON               |
| ON               | ON               | ON               | ON           | ON           | ON         | ON         | ON               |

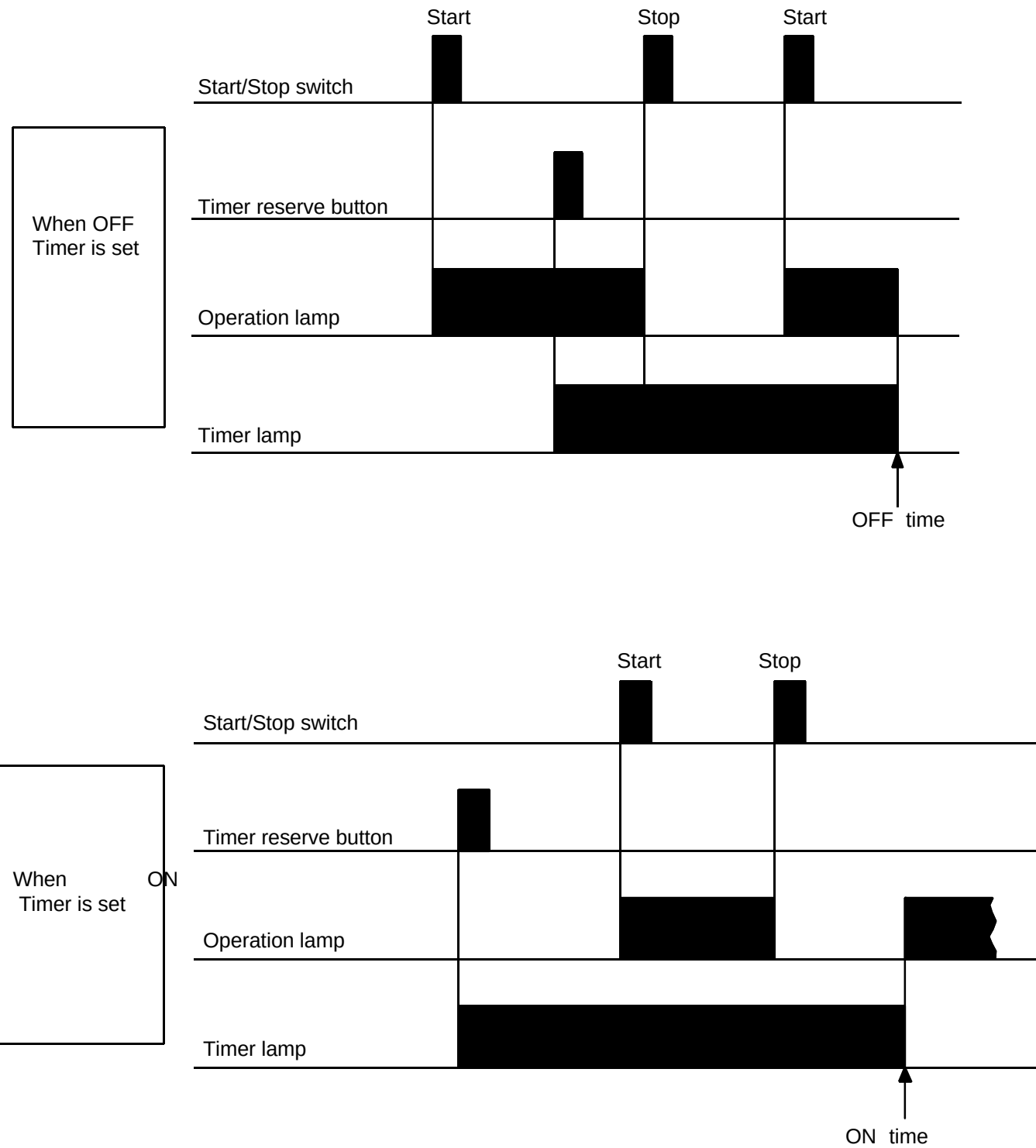
# AUTO SWING FUNCTION

| INPUT SIGNAL                      |                  | PRESENT CONDITION                           |                          | OPERATING SPECIFICATION                                                                                     | REFERENCE                     |
|-----------------------------------|------------------|---------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------|
|                                   |                  | OPERATION                                   | AIR DEFLECTOR            |                                                                                                             |                               |
| KEY INPUT                         | STOP             | EACH MODE                                   | STOP                     | ONE SWING (CLOSING AIR DEFLECTOR)<br>DOWNWARD<br>UPWARD                                                     | INITIALIZE AT NEXT OPERATION. |
|                                   |                  |                                             | DURING ONE SWING         | STOP AT THE MOMENT.                                                                                         |                               |
|                                   | DURING OPERATION | AUTO COOL<br>COOL<br>FAN<br>AUTO DRY<br>DRY | STOP                     | START SWINGING<br>DOWNWARD<br>UPWARD<br>DOWNWARD                                                            |                               |
|                                   |                  |                                             | DURING SWING             | STOP AT THE MOMENT.                                                                                         |                               |
|                                   |                  | CIRCULATOR                                  | STOP                     | START SWINGING<br>DOWNWARD<br>UPWARD<br>DOWNWARD                                                            |                               |
|                                   |                  |                                             | DURING SWING             | STOP AT THE MOMENT.                                                                                         |                               |
| THERMO. (INTERNAL ON)<br>ON FAN   | DURING OPERATION | AUTO DRY<br>DRY<br>CIRCULATOR               | TEMPORARY STOP           | START SWING AGAIN.                                                                                          |                               |
| THERMO. (INTERNAL OFF)<br>OFF FAN |                  |                                             | DURING SWINGING          | STOP SWINGING TEMPORARILY<br>(SWING MODE IS CLEARED IF SWING COMMAND IS TRANSMITTED DURING TEMPORARY STOP.) |                               |
| MAIN SWITCH<br>ON                 | STOP             | COOL<br>FAN<br>DRY                          | STOP<br>DURING ONE SWING | INITIALIZE<br>DOWNWARD<br>UPWARD                                                                            |                               |
|                                   |                  | CIRCULATOR                                  | STOP<br>DURING ONE SWING | INITIALIZE<br>DOWNWARD                                                                                      |                               |
| MAIN SWITCH<br>OFF                | DURING OPERATION | EACH MODE                                   | STOP<br>DURING SWING     | ONE SWING (CLOSING AIR DEFLECTOR)<br>DOWNWARD<br>UPWARD                                                     | INITIALIZE AT NEXT OPERATION. |
|                                   |                  |                                             | DURING INITIALIZING      |                                                                                                             |                               |
| CHANGE OF OPERATION               | DURING OPERATION | EACH MODE                                   | STOP                     | INITIALIZING CONDITION OF EACH MODE.                                                                        |                               |
|                                   |                  |                                             | DURING SWING             | STOP SWINGING AND MODE BECOMES INITIALIZING CONDITION.                                                      |                               |

**DESCRIPTION OF MAIN CIRCUIT OPERATION**

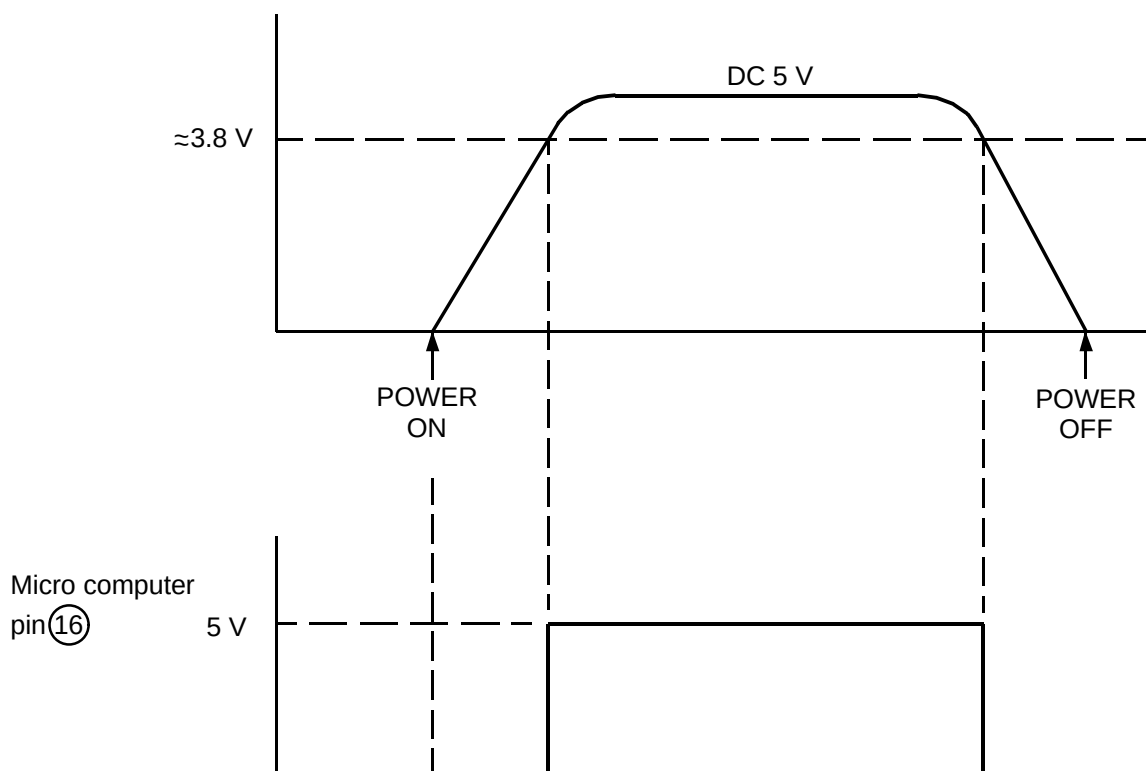
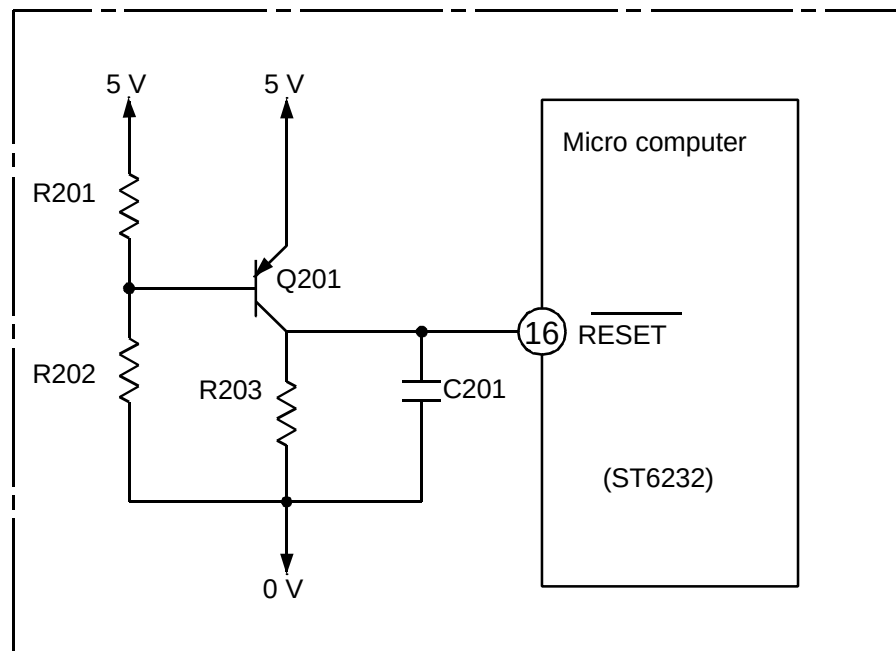
**1. ON/OFF**

The Start/Stop and Timer reserve button independently. Their operations are shown in Fig. 1.



**Fig. 1 Timer operation**

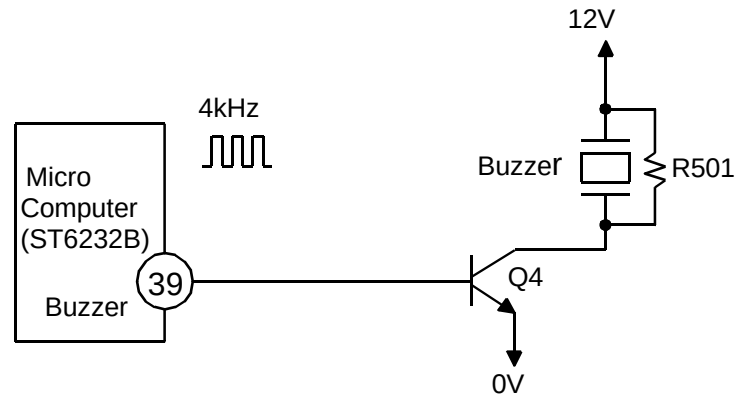
## 2.Reset circuit



- The reset circuit is used to reset the program to its initial settings when the power is turned on or when the power is recovered after a power failure.
- The micro computer is reset when the reset input is “Lo”, and operation is possible when the reset input is “Hi”.
- The waveforms at each point when the power is turned on and off are shown in the diagrams.

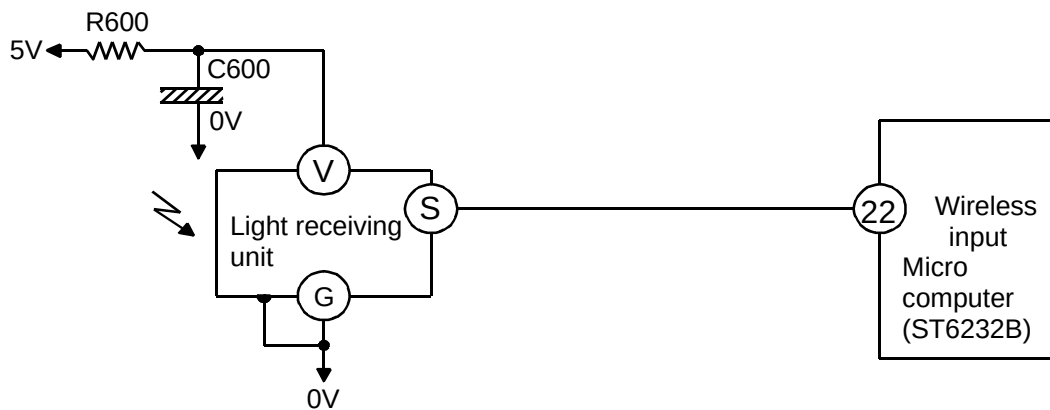


### 3. Buzzer circuit



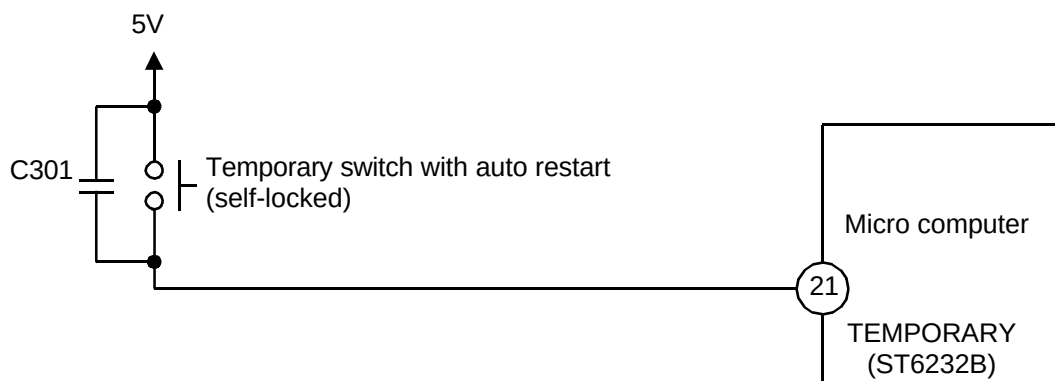
When the buzzer is to be activated, buzzer output pin 39 of the micro computer alternates between ON and OFF repeatedly at 4kHz. A 4kHz voltage is applied to the buzzer and the diaphragm of the buzzer vibrates to output 4kHz sound.

### 4. Receive circuit



Infrared signals from the wireless remote controller are received by the light receiving unit and output after being amplified and shaped.

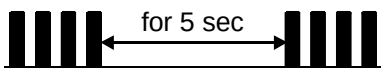
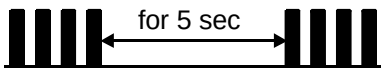
### 5. Temporary Switch Circuit with Auto Restart function.



The temporary switch with Auto Restart function is used when the wireless remote controller is lost or troubled, or when power is supplied again after power stoppage.

[Self-diagnosis from indoor indicator lamp]

Check the position of the problem judging from the number of blinks of indoor indicator lamp.

| Blinking state of the timer lamp                                                                        | Details of operation                                                                                                                    | Parts to be checked                               |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <p>Blinks 14times</p>  | <p>When the indoor unit heat exchanger thermistor is short-circuited or disconnected, all operations will stop. (Reset to recover.)</p> | <p>(1) Check the heat exchanger thermistor.</p>   |
| <p>Blinks 4 times</p>  | <p>When the room temperature thermistor is short-circuited or disconnected, all operations will stop. (Reset to recover.)</p>           | <p>(1) Check the room temperature thermistor.</p> |

Note:

When the timer lamp is blinking, operation using the remote controller cannot be done. To check the operation again, turn the power switch off, then turn it on again.

## SERVICE CALL Q&A

### Cooling operation

**Q1** While cooling, the compressor sometimes stops abruptly.



**A1** Check whether frost sticks on the heat exchanger of indoor unit or not. Wait for 3-4 minutes until the frost melts.

If cooling is performed when the room temperature is low, frost may stick on the heat exchanger of indoor unit.

### Dehumidifying operation

**Q2** The fan speed does not change during a dehumidifying operation.



**A2** The fan speed is always LO at a dehumidifying operation.

**Q3** Cold air comes out during a dehumidifying operation.



**A3** To improve the dehumidification efficiency, LO fan speed operation is performed. Therefore the air is cold. This is not a trouble.

**Q4** The operation does not stop even by raising the room temperature setting of remote control at a dehumidifying operation.



**A4** At a dehumidifying operation, the actual room temperature is compared with the room temperature setting when starting the operation and the operation is as follows.

- 1) When actual room temperature  $\geq$  room temperature setting +1. The operation is according to the room temperature setting on the remote controller.
- 2) When actual room temperature < room temperature setting +1. Regardless of the room temperature setting, the temperature is automatically set = room temp. -2. In this case, the status is as 2) and therefore, the operation by the room temperature control is impossible. Turn off the ON/OFF switch, set the room temperature to a new value and turn on the operation by the ON/OFF switch.

**Q5** In the dehumidifying mode, the temperature set by remote controller is set slightly higher than the room temperature but the operation starts.



**A5** This is the status in 2) of (A4). The temperature is set a little lower than the room temperature to carry out a dehumidifying operation as far as possible.

## Automatic operation

**Q6** How is the automatic operation mode determined?



**A6** According to the room temperature, cooling or dehumidifying operation is automatically selected.  
Cooling:  
When room temperature is approx. 25 or higher  
Dehumidify:  
When room temperature is between approx. 16 and 25

**Q7** At an automatic operation, changing the fan speed change switch does not vary the fan speed.



**A7** The fan speed is automatically determined.

**Q8** The setting temperature cannot be controlled at an automatic operation.



**A8** It is automatically set as follows.  
  
At cooling :Set at 25  
At dehumidifying:  
If  $RT \geq 24$   
then ST 23  
else ST RT-2  
  
RT: Room temperature  
ST: Setting temperature

When changing the room temperature setting in an automatic operation, the next automatic operation mode is determined by new room temperature setting. If, for example,

Cooling : When room Temperature is Lower Than 22

Dehumidifying:  
When room temperature is higher than 26

Common, etc.

**Q9** There is a difference between the room temperature setting and actual room temperature.



**A9** There may be a difference between the room temperature setting and actual room temperature on account of the room structure, air flow, etc.  
If there is a difference from the room temperature, adjust the set temperature to keep living space at a comfortable

**Q10** What will happen if the time setting is changed while in a timer operation?



**A10** A timer operation is performed until the time after changing the time setting.

1) When the Automatic

**Q11** In the "Automatic fan speed" mode, the indoor fan changes to MED and LO fan speed.



2) When the operation

**A11** It is not a trouble. The cold wind preventive function operates.

setting is not displayed.

Wireless remote controller

**Q12**

3) Pressing the room temperature control button  transmits mark " " and sounds a receive sound but does not display the room temperature setting.



displayed.

The room temperature is automatically set as follows.

At cooling

Set at 25

At dehumidifying

Set to a temperature slightly lower than the room temperature.

3) At an "Automatic" operation, the room temperature setting is not displayed.

The room temperature sensing thermistor in the indoor unit detects the room temperature and, according to the particular temperature, automatically performs "Cooling", or Dehumidifying" operation. The value indicated not at an "Automatic" but manual operation is not the actual room temperature but the room temperature setting.

**Q13** When the room temperature setting is "16", pressing the room temperature control button "V" causes no transmission. At "32", pressing "E" causes no transmission either.

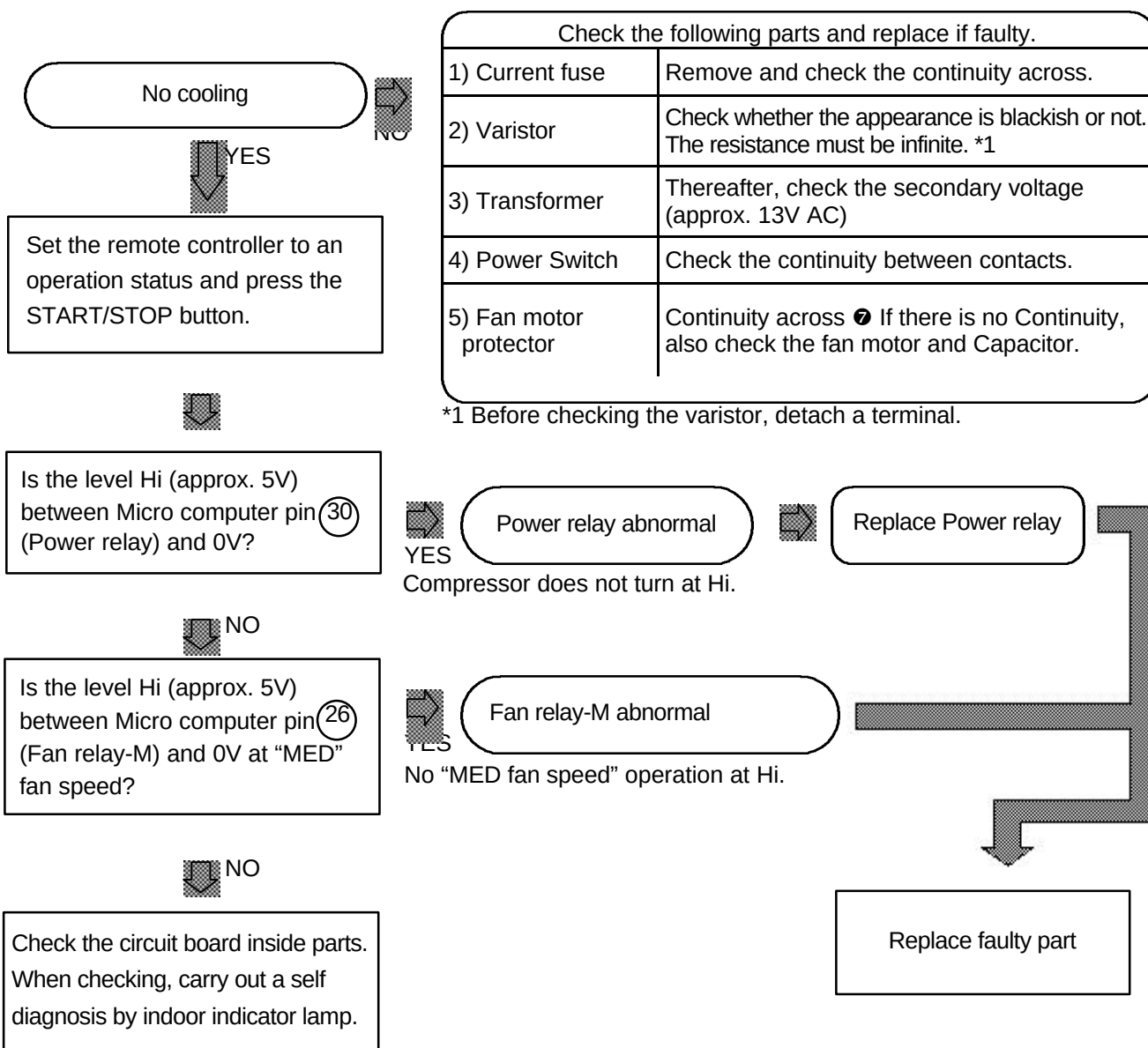


**A13** The room temperature is settable within the range of 16-32 and not beyond.

**Q14** After selecting a "dehumidifying" operation mode, the fan speed mode remains "LO fan speed".



**A14** At a "dehumidifying" operation, the "LO fan speed" is forcibly selected.

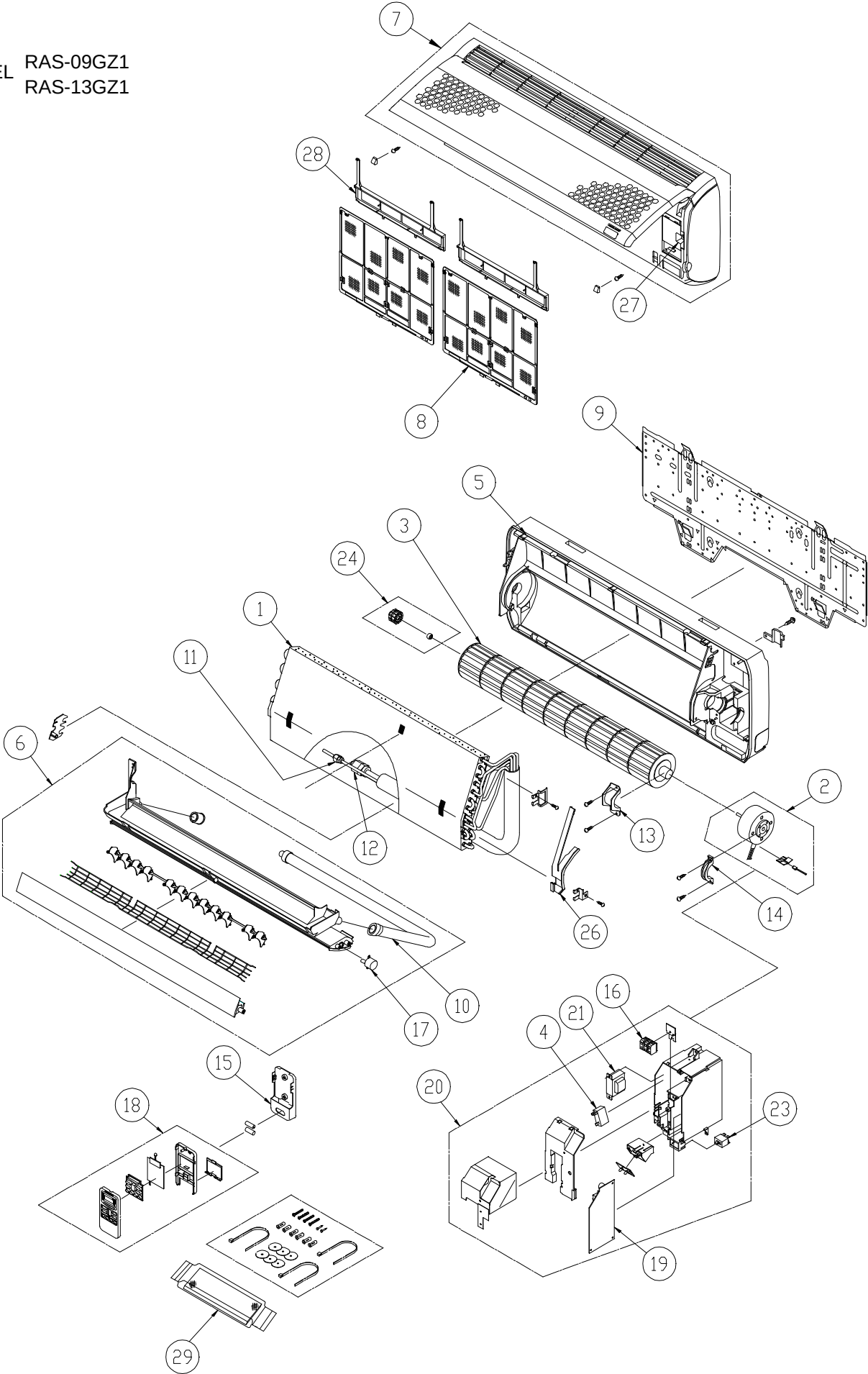


### [Self-diagnosis check method with the unit indication lamp]

| Trouble mode                                                         | Indication lamp      | Operation contents                                                                                                            | Inspection parts                                      |
|----------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Defective operation of the room temperature thermistor or defroster. | Timer lamp flashing. | Full stop occurs with a short-circuit or a wire break of the room temperature thermistor or defroster. ( Reset for recovery ) | Inspect the room temperature thermistor or defroster. |

PARTS LIST AND DIAGRAM

MODEL RAS-09GZ1  
RAS-13GZ1





MODEL RAS-09GZ1

| NO. | PARTS NO.<br>RAS-09GZ1 | Q'TY/UNIT | PARTS NAME                 | REMARK |
|-----|------------------------|-----------|----------------------------|--------|
| 1   | RAS-09GZ1 601          | 1         | EVAPORATOR                 |        |
| 2   | RAS-09GZ1 602          | 1         | FAN MOTOR (8W)             |        |
| 3   | RAS-09GZ1 603          | 1         | TANGENTIAL FLOW FAN        |        |
| 4   | RAS-09GZ1 604          | 1         | CAPACITOR(1 $\mu$ F,450V)  |        |
| 5   | RAS-09GZ1 605          | 1         | CABINET                    |        |
| 6   | RAS-09GZ1 606          | 1         | DRAIN PAN ASSEMBLY         |        |
| 7   | RAS-09GZ1 607          | 1         | FRONT COVER ASSEMBLY       |        |
| 8   | RAS-09GZ1 608          | 2         | FILTER                     |        |
| 9   | RAS-09GZ1 609          | 1         | MOUNTING PLATE             |        |
| 10  | RAS-09GZ1 610          | 1         | DRAIN HOSE                 |        |
| 11  | RAS-09GZ1 611          | 1         | UNION(2)                   |        |
| 12  | RAS-09GZ1 612          | 1         | UNION(3)                   |        |
| 13  | RAS-09GZ1 613          | 1         | FAN MOTOR SUPPORT(L)       |        |
| 14  | RAS-09GZ1 614          | 1         | FAN MOTOR SUPPORT(R)       |        |
| 15  | RAS-09GZ1 615          | 1         | REMOTE CONTROLLER SUPPORT  |        |
| 16  | RAS-09GZ1 616          | 1         | TERMINAL BOARD(3P)         |        |
| 17  | RAS-09GZ1 617          | 1         | STEPPING MOTOR             |        |
| 18  | RAS-09GZ1 618          | 1         | REMOTE CONTROLLER          |        |
| 19  | RAS-09GZ1 619          | 1         | P.C.B.(MAIN)               |        |
| 20  | RAS-09GZ1 620          | 1         | CONTROL BODY(MIQ-RES07NB1) |        |
| 21  | RAS-09GZ1 621          | 1         | TRANSFORMER                |        |
| 23  | RAS-09GZ1 623          | 1         | POWER SWITCH               |        |
| 24  | RAS-09GZ1 624          | 1         | BEARING ASSEMBLY           |        |
| 26  | RAS-09GZ1 626          | 1         | PIPE COVER                 |        |
| 27  | RAS-09GZ1 627          | 1         | COVER                      |        |
| 28  | RAS-09GZ1 628          | 2         | FILTER HOLDER              |        |
| 29  | RAS-09GZ1 629          | 1         | AIR CLEANING FILTER        |        |

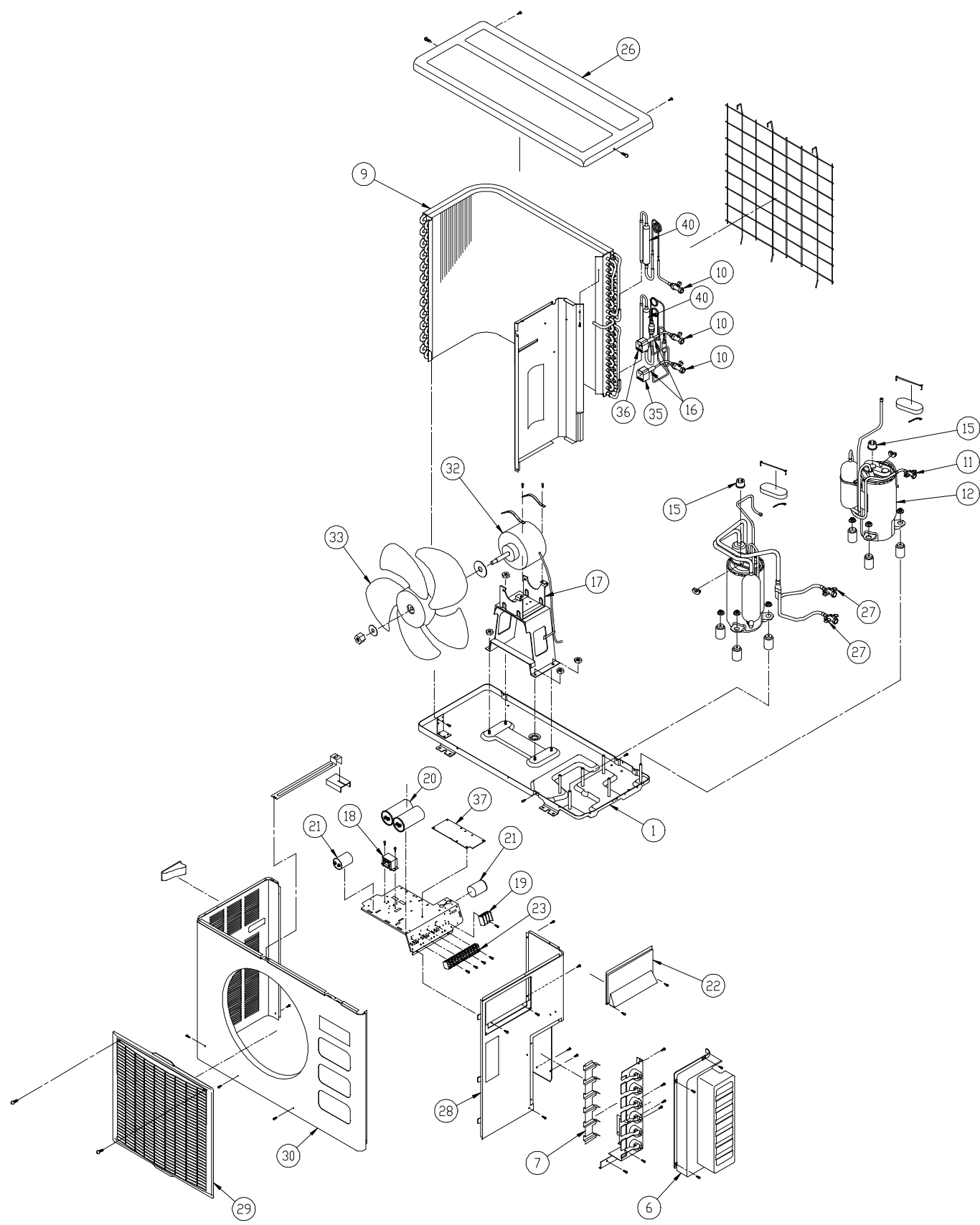
| NO. | PARTS NO.<br>RAS-09GZ1 | Q'TY/UNIT | PARTS NAME                      | REMARK |
|-----|------------------------|-----------|---------------------------------|--------|
| 31  | RAS-09GZ1 631          | 1         | P.C.B (LED)                     |        |
| 32  | RAS-09GZ1 632          | 1         | TRANSFORMER                     |        |
| 33  | RAM-20CW1 616          | 1         | RELAY (JM1a-TMP-DC12V)          |        |
| 34  | RAS-09GZ1 633          | 1         | THERMISTOR (ROOM TEMP.)         |        |
| 35  | RAS-09GZ1 634          | 1         | FUSE (2A)                       |        |
| 36  | RAM-20CW1 612          | 2         | FUSE HOLDER                     |        |
| 37  | RAS-09GZ1 635          | 1         | OSCILLATOR (CST6.00MGW)         |        |
| 38  | RAM-20CW1 613          | 2         | VARISTOR (470NR)                |        |
| 39  | RAS-09GZ1 636          | 2         | TRANSISTOR (2SA1015)            |        |
| 40  | RAS-09GZ1 637          | 1         | TRANSISTOR (DTA114TS)           |        |
| 41  | RAS-09GZ1 638          | 2         | TRANSISTOR (DTC114TS)           |        |
| 42  | RAS-09GZ1 639          | 1         | MICRO COMPUTER (ST6232B)        |        |
| 43  | RAM-20CW1 626          | 1         | SURGE ABSORBER(DSA-362MA)       |        |
| 44  | RAM-20CW1 624          | 3         | DIODE (IN4003)                  |        |
| 45  | RAS-09GZ1 640          | 1         | DIODE BRIDGE (SIVB20)           |        |
| 46  | RAS-09GZ1 641          | 1         | LED (YELLOW)(L-44YD-LASF2.5)    |        |
| 47  | RAS-09GZ1 642          | 1         | LED (GREEN)(L-44GD-LASF2.5)     |        |
| 48  | RW-123BHS 203          | 1         | POST (VH-PV-2PB)                |        |
| 49  | RAS-09GZ1 643          | 1         | CAPACITOR(0.01 $\mu$ F, 50V)    |        |
| 50  | RAS-09GZ1 644          | 1         | REGULATOR IC (78M05)            |        |
| 51  | RAM-20CW1 618          | 1         | REGULATOR IC ( $\mu$ PC7812AHF) |        |
| 52  | RAS-09GZ1 645          | 1         | LED COVER                       |        |
| 53  | RAS-09GZ1 646          | 1         | LIGHT RECEIVING UNIT            |        |
| 54  | RAS-09GZ1 647          | 1         | FAN MOTOR PROTECTOR (115 )      |        |
| 55  | RAS-09GZ1 648          | 1         | THERMISTOR (HEAT EXCHANGER)     |        |
| 56  | RAS-5101C 916          | 1         | DRIVER-IC (ULN2003AN)           |        |

MODEL RAS-13GZ1

| NO. | PARTS NO.<br>RAS-13GZ1 | Q'TY/UNIT | PARTS NAME                  | REMARK |
|-----|------------------------|-----------|-----------------------------|--------|
| 1   | RAS-13GZ1 601          | 1         | EVAPORATOR                  |        |
| 2   | RAS-13GZ1 602          | 1         | FAN MOTOR (10W)             |        |
| 3   | RAS-09GZ1 603          | 1         | TANGENTIAL FLOW FAN         |        |
| 4   | RAS-09GZ1 604          | 1         | CAPACITOR (1 $\mu$ F,450V)  |        |
| 5   | RAS-09GZ1 605          | 1         | CABINET                     |        |
| 6   | RAS-09GZ1 606          | 1         | DRAIN PAN ASSEMBLY          |        |
| 7   | RAS-09GZ1 607          | 1         | FRONT COVER ASSEMBLY        |        |
| 8   | RAS-09GZ1 608          | 2         | FILTER                      |        |
| 9   | RAS-09GZ1 609          | 1         | MOUNTING PLATE              |        |
| 10  | RAS-09GZ1 610          | 1         | DRAIN HOSE                  |        |
| 11  | RAS-09GZ1 611          | 1         | UNION(2)                    |        |
| 12  | RAS-13GZ1 603          | 1         | UNION(4)                    |        |
| 13  | RAS-09GZ1 613          | 1         | FAN MOTOR SUPPORT(L)        |        |
| 14  | RAS-09GZ1 614          | 1         | FAN MOTOR SUPPORT(R)        |        |
| 15  | RAS-09GZ1 615          | 1         | REMOTE CONTROLLER SUPPORT   |        |
| 16  | RAS-09GZ1 616          | 1         | TERMINAL BOARD(3P)          |        |
| 17  | RAS-09GZ1 617          | 1         | STEPPING MOTOR              |        |
| 18  | RAS-09GZ1 618          | 1         | REMOTE CONTROLLER           |        |
| 19  | RAS-09GZ1 619          | 1         | P.C.B.(MAIN)                |        |
| 20  | RAS-09GZ1 620          | 1         | CONTROL BODY (MIQ-RES07NB1) |        |
| 21  | RAS-09GZ1 621          | 1         | TRANSFORMER                 |        |
| 23  | RAS-09GZ1 623          | 1         | POWER SWITCH                |        |
| 24  | RAS-09GZ1 624          | 1         | BEARING ASSEMBLY            |        |
| 26  | RAS-09GZ1 626          | 1         | PIPE COVER                  |        |
| 27  | RAS-09GZ1 627          | 1         | COVER                       |        |
| 28  | RAS-09GZ1 628          | 2         | FILTER HOLDER               |        |
| 29  | RAS-09GZ1 629          | 1         | AIR CLEANING FILTER         |        |

| NO. | PARTS NO.<br>RAS-13GZ1 | Q'TY/UNIT | PARTS NAME                      | REMARK |
|-----|------------------------|-----------|---------------------------------|--------|
| 31  | RAS-09GZ1 631          | 1         | P.C.B (LED)                     |        |
| 32  | RAS-09GZ1 632          | 1         | TRANSFORMER                     |        |
| 33  | RAM-20CW1 616          | 1         | RELAY (JM1a-TMP-DC12V)          |        |
| 34  | RAS-09GZ1 633          | 1         | THERMISTOR (ROOM TEMP.)         |        |
| 35  | RAS-09GZ1 634          | 1         | FUSE (2A)                       |        |
| 36  | RAM-20CW1 612          | 2         | FUSE HOLDER                     |        |
| 37  | RAS-09GZ1 635          | 1         | OSCILLATOR (CST6.00MGW)         |        |
| 38  | RAM-20CW1 613          | 2         | VARISTOR (470NR)                |        |
| 39  | RAS-09GZ1 636          | 2         | TRANSISTOR (2SA1015)            |        |
| 40  | RAS-09GZ1 637          | 1         | TRANSISTOR (DTA114TS)           |        |
| 41  | RAS-09GZ1 638          | 2         | TRANSISTOR (DTC114TS)           |        |
| 42  | RAS-09GZ1 639          | 1         | MICRO COMPUTER (ST6232B)        |        |
| 43  | RAM-20CW1 626          | 1         | SURGE ABSORBER(DSA-362MA)       |        |
| 44  | RAM-20CW1 624          | 3         | DIODE (IN4003)                  |        |
| 45  | RAS-09GZ1 640          | 1         | DIODE BRIDGE (SIVB20)           |        |
| 46  | RAS-09GZ1 641          | 1         | LED (YELLOW)(L-44YD-LASF2.5)    |        |
| 47  | RAS-09GZ1 642          | 1         | LED (GREEN)(L-44GD-LASF2.5)     |        |
| 48  | RW-123BHS 203          | 1         | POST (VH-PV-2PB)                |        |
| 49  | RAS-09GZ1 643          | 1         | CAPACITOR(0.01 $\mu$ F, 50V)    |        |
| 50  | RAS-09GZ1 644          | 1         | REGULATOR IC (78M05)            |        |
| 51  | RAM-20CW1 618          | 1         | REGULATOR IC ( $\mu$ PC7812AHF) |        |
| 52  | RAS-09GZ1 645          | 1         | LED COVER                       |        |
| 53  | RAS-09GZ1 646          | 1         | LIGHT RECEIVING UNIT            |        |
| 54  | RAS-09GZ1 647          | 1         | FAN MOTOR PROTECTOR (115 )      |        |
| 55  | RAS-09GZ1 648          | 1         | THERMISTOR (HEAT EXCHANGER)     |        |
| 56  | RAS-5101C 916          | 1         | DRIVER-IC (ULN2003AN)           |        |

MODEL RAM-20GW1

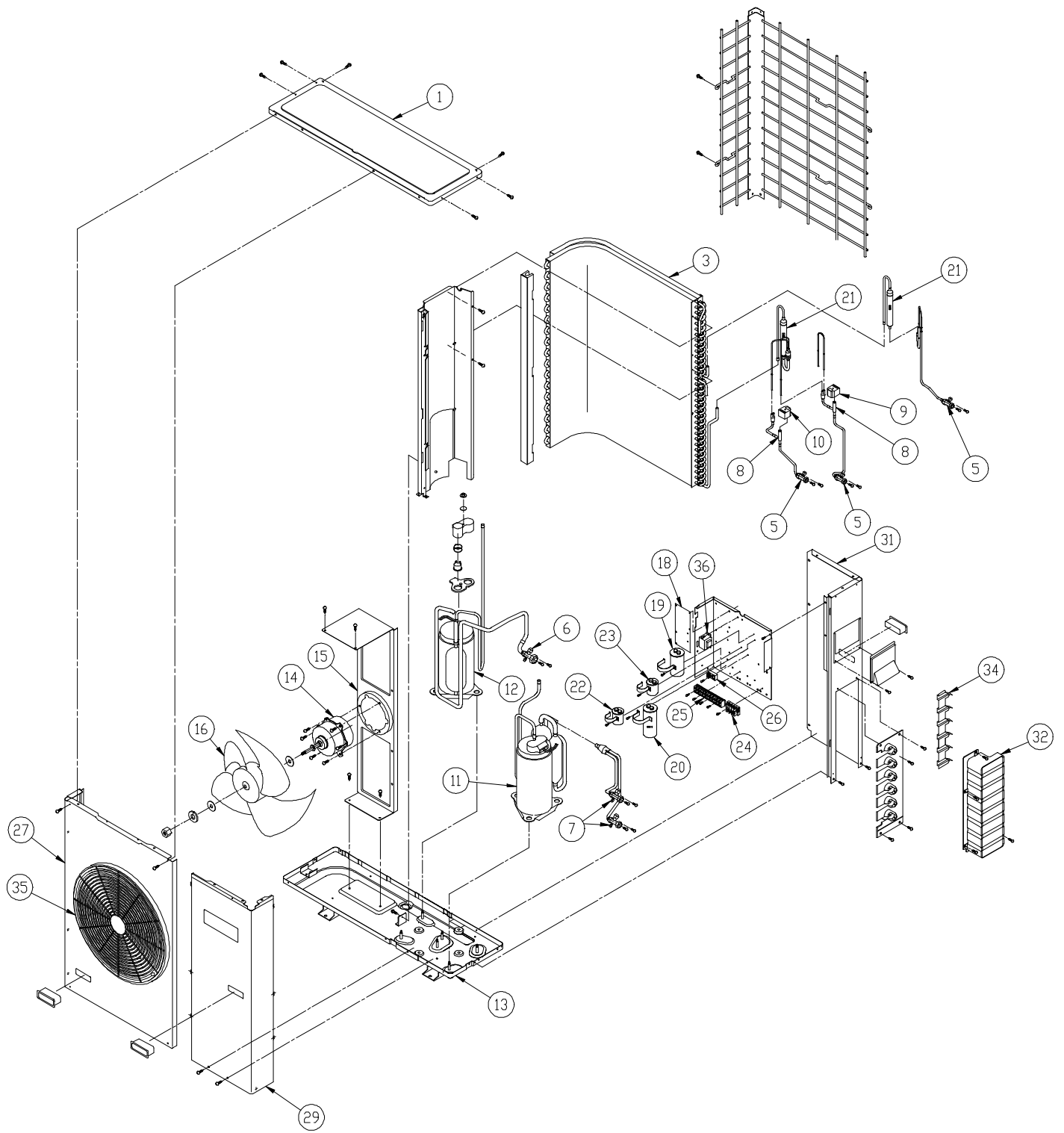


MODEL RAM-20GW1

| NO. | PARTS NO.<br>RAS-20GW1 | Q'TY/UNIT | PARTS NAME                      | REMARK |
|-----|------------------------|-----------|---------------------------------|--------|
| 1   | RAM-18GZ1 601          | 1         | BASE                            |        |
| 6   | RAM-20CW1 601          | 1         | VALVE COVER                     |        |
| 7   | RAM-20CW1 602          | 1         | RUBBER                          |        |
| 9   | RAM-20GW1 602          | 1         | CONDENSER                       |        |
| 10  | RAM-20GW1 603          | 3         | 2-WAY VALVE ( 1/4" )            |        |
| 11  | RAM-18GZ1 606          | 1         | 3-WAY VALVE ( 3/8" )            |        |
| 12  | RAM-18GZ1 607          | 2         | COMPRESSOR ( GK516S1Z )         |        |
| 15  | RAM-18GZ1 620          | 2         | OVERLOAD RELAY ( 8.0S66C30A )   |        |
| 16  | RAM-20CW1 605          | 2         | 2-WAY VALVE                     |        |
| 17  | RAM-18CZ1 619          | 1         | MOTOR BASE                      |        |
| 18  | RAM-20CW1 606          | 1         | TRANSFORMER                     |        |
| 19  | RAM-18CHZ1 605         | 1         | TERMINAL BOARD ( 3P )           |        |
| 20  | RAM-18GZ1 610          | 2         | CAPACITOR ( 40 $\mu$ F, 400V )  |        |
| 21  | RAM-18CZ1 611          | 2         | CAPACITOR ( 2.3 $\mu$ F, 450V ) |        |
| 22  | RAM-18GZ1 623          | 1         | ELECTRICAL COVER                |        |
| 23  | RAM-30CW1 613          | 4         | TERMINAL BOARD ( 3P )           |        |
| 26  | RAM-18GZ1 613          | 1         | TOP COVER                       |        |
| 27  | RAM-20GW1 604          | 2         | 3-WAY VALVE ( 3/8" )            |        |
| 28  | RAM-20GW1 607          | 1         | SIDE COVER                      |        |
| 29  | RAM-18GZ1 615          | 1         | SPIDER                          |        |
| 30  | RAM-18GZ1 616          | 1         | CABINET                         |        |
| 32  | RAM-20GW1 608          | 1         | FAN MOTOR ( 40W )               |        |
| 33  | RAM-18CZ1 618          | 1         | PROPELLOR FAN                   |        |
| 35  | RAM-20CW1 609          | 1         | COIL ( FOR 2-WAY VALVE )        |        |
| 36  | RAM-20CW1 610          | 1         | COIL ( FOR 2-WAY VALVE )        |        |
| 37  | RAM-20CW1 635          | 1         | P.W.B.                          |        |
| 40  | RAM-18GZ1 634          | 2         | DRYER                           |        |
| 77  | RAM-20CW1 611          | 1         | FUSE ( 2.5A )                   |        |
| 78  | RAM-20CW1 612          | 1         | FUSE HOLDER                     |        |
| 79  | RAM-20CW1 613          | 2         | VARISTOR ( 470NR )              |        |
| 80  | RAM-20CW1 614          | 3         | PHOTO COUPLER ( PS-2501 )       |        |



MODEL RAM-30GW1





MODEL RAM-30GW1

| NO. | PARTS NO.<br>RAS-30GW1 | Q'TY/UNIT | PARTS NAME                      | REMARK |
|-----|------------------------|-----------|---------------------------------|--------|
| 1   | RAM-30CW1 601          | 1         | TOP COVER                       |        |
| 3   | RAM-30GW1 602          | 1         | CONDENSER                       |        |
| 5   | RAM-20GW1 603          | 3         | 2-WAY VALVE ( 1/4" )            |        |
| 6   | RAM-30GW1 603          | 1         | 3-WAY VALVE ( 1/2" )            |        |
| 7   | RAM-20GW1 604          | 2         | 3-WAY VALVE ( 3/8" )            |        |
| 8   | RAM-20CW1 605          | 2         | 2-WAY VALVE                     |        |
| 9   | RAM-20CW1 609          | 1         | COIL ( FOR 2-WAY VALVE )        |        |
| 10  | RAM-30GW1 608          | 1         | COIL ( FOR 2-WAY VALVE )        |        |
| 11  | RAM-30GW1 604          | 1         | COMPRESSOR ( 4KS340DAA )        |        |
| 12  | RAM-30GW1 605          | 1         | COMPRESSOR ( 4KS210DAA )        |        |
| 13  | RAM-30CW1 606          | 1         | BASE                            |        |
| 14  | RAM-30GW1 607          | 1         | FAN MOTOR ( 75W )               |        |
| 15  | RAM-30CW1 608          | 1         | MOTOR BASE                      |        |
| 16  | RAM-30CW1 609          | 1         | PROPELLER FAN                   |        |
| 18  | RAM-20CW1 635          | 1         | P.W.B.                          |        |
| 19  | RAM-30GW1 609          | 1         | CAPACITOR ( 35 $\mu$ F, 400V )  |        |
| 20  | RAM-30GW1 610          | 1         | CAPACITOR ( 60 $\mu$ F, 400V )  |        |
| 21  | RAM-18GZ1 634          | 2         | DRYER                           |        |
| 22  | RAM-30CW1 611          | 1         | CAPACITOR ( 4 $\mu$ F, 500V )   |        |
| 23  | RAM-30CW1 612          | 1         | CAPACITOR ( 2.3 $\mu$ F, 400V ) |        |
| 24  | RAM-30GW1 611          | 1         | TERMINAL BOARD( 3P )            |        |
| 25  | RAM-30CW1 613          | 4         | TERMINAL BOARD( 3P )            |        |
| 26  | RAM-30CW1 614          | 1         | TERMINAL BOARD                  |        |
| 27  | RAM-30CW1 615          | 1         | CABINET                         |        |
| 29  | RAM-30CW1 616          | 1         | SERVICE COVER                   |        |
| 31  | RAM-30CW1 617          | 1         | SIDE COVER                      |        |
| 32  | RAM-30CW1 618          | 1         | VALVE COVER                     |        |
| 34  | RAM-20CW1 602          | 1         | RUBBER                          |        |
| 35  | RAM-30CW1 619          | 1         | SPIDER                          |        |
| 36  | RAM-20CW1 606          | 1         | TRANSFORMER                     |        |

| NO. | PARTS NO.<br>RAS-30GW1 | Q'TY/UNIT | PARTS NAME                      | REMARK |
|-----|------------------------|-----------|---------------------------------|--------|
| 77  | RAM-20CW1 611          | 1         | FUSE ( 2.5A )                   |        |
| 78  | RAM-20CW1 612          | 1         | FUSE HOLDER                     |        |
| 79  | RAM-20CW1 613          | 2         | VARISTOR ( 470NR )              |        |
| 80  | RAM-20CW1 614          | 3         | PHOTO COUPLER ( PS-2501 )       |        |
| 81  | RAM-20CW1 615          | 4         | DIODE BRIDGE ( S1WB60 )         |        |
| 82  | RAM-20CW1 616          | 2         | RELAY ( JM1aN-TMP-DC12V )       |        |
| 83  | RAM-20CW1 617          | 1         | RELAY ( JS1-12V )               |        |
| 84  | RAM-20CW1 618          | 1         | REGULATOR IC ( 7812 )           |        |
| 85  | RAM-20CW1 619          | 1         | REGULATOR IC ( 7805 )           |        |
| 86  | RAM-20CW1 620          | 1         | TIMER IC ( AN6780 )             |        |
| 87  | RAM-20CW1 621          | 1         | LOGIC IC ( 7432 )               |        |
| 88  | RAM-20CW1 622          | 1         | LOGIC IC ( 7402 )               |        |
| 89  | RAM-20CW1 623          | 1         | DRIVER IC ( 2003 )              |        |
| 90  | RAM-20CW1 624          | 2         | DIODE ( IN4003 )                |        |
| 91  | RAM-20CW1 625          | 1         | SPARK KILLER ( XE1201 )         |        |
| 92  | RAM-20CW1 626          | 1         | SURGE ABSORBER ( S-ABSO-362 )   |        |
| 94  | RAM-20CW1 627          | 1         | SOLID STATE RELAY ( S26MD02V )  |        |
| 95  | RAM-20CW1 628          | 1         | CAPACITOR ( 1000 $\mu$ F, 35V ) |        |
| 96  | RAM-20CW1 629          | 4         | CAPACITOR ( 0.1 $\mu$ F, 25V )  |        |
| 97  | RAM-20CW1 630          | 2         | CAPACITOR ( 0.1 $\mu$ F, 50V )  |        |
| 98  | RAM-20CW1 631          | 1         | CAPACITOR ( 100 $\mu$ F, 50V )  |        |
| 99  | RAM-20CW1 632          | 3         | CAPACITOR ( 10 $\mu$ F, 16V )   |        |
| 100 | RAM-20CW1 633          | 3         | CAPACITOR ( 47 $\mu$ F, 16V )   |        |
| 101 | RAM-20CW1 634          | 2         | CAPACITOR ( 1000PF, AC250V )    |        |



---

RAS-09GZ1 × 3 / RAM-20GW1  
RAS-13GZ1  
RAS-09GZ1×2 / RAM-30GW1

Printed in TAIWAN R.O.C.